

BFS Traversal

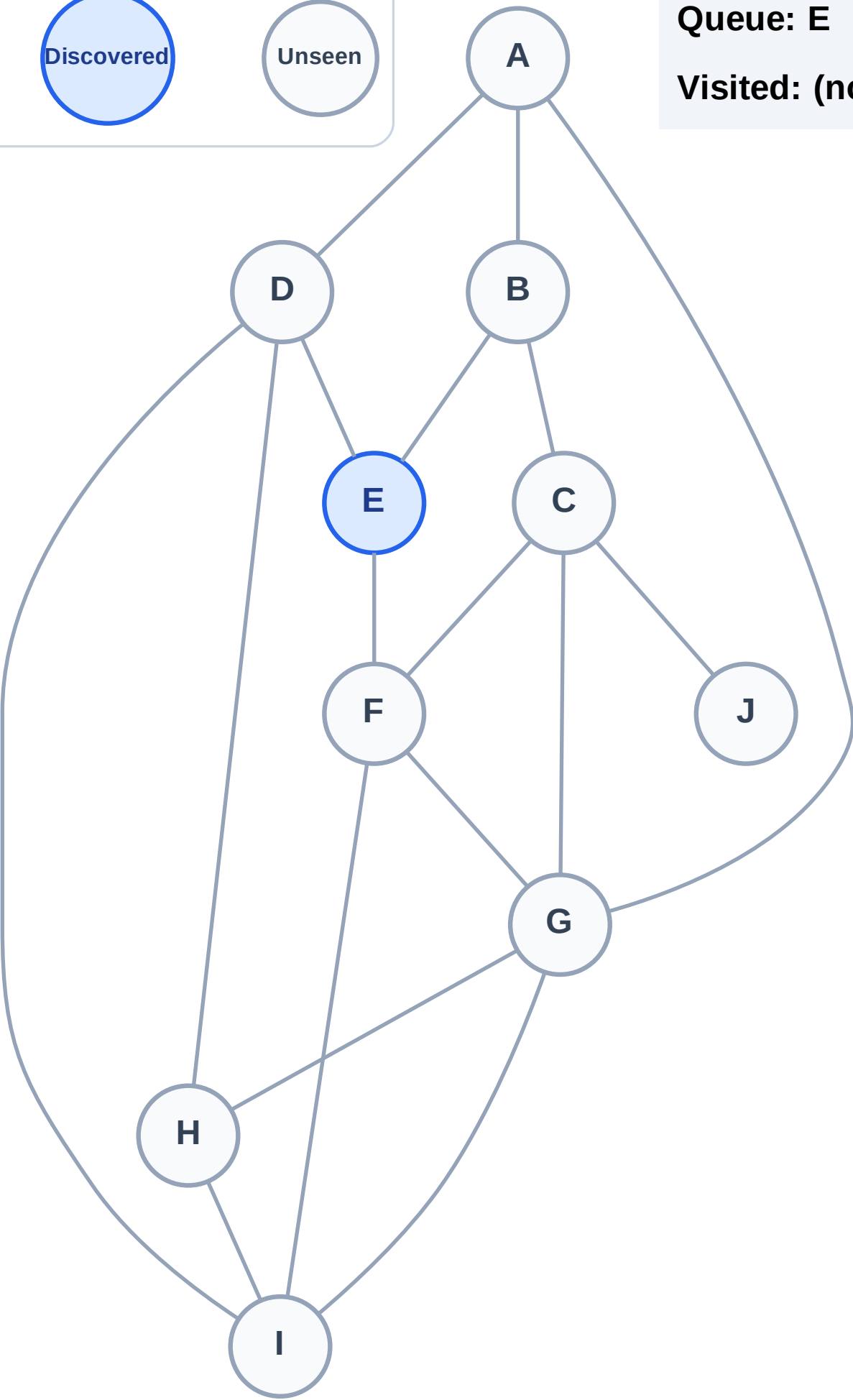
Step 1: Start at E

Legend



Queue: E

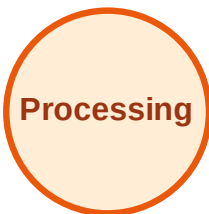
Visited: (none)



BFS Traversal

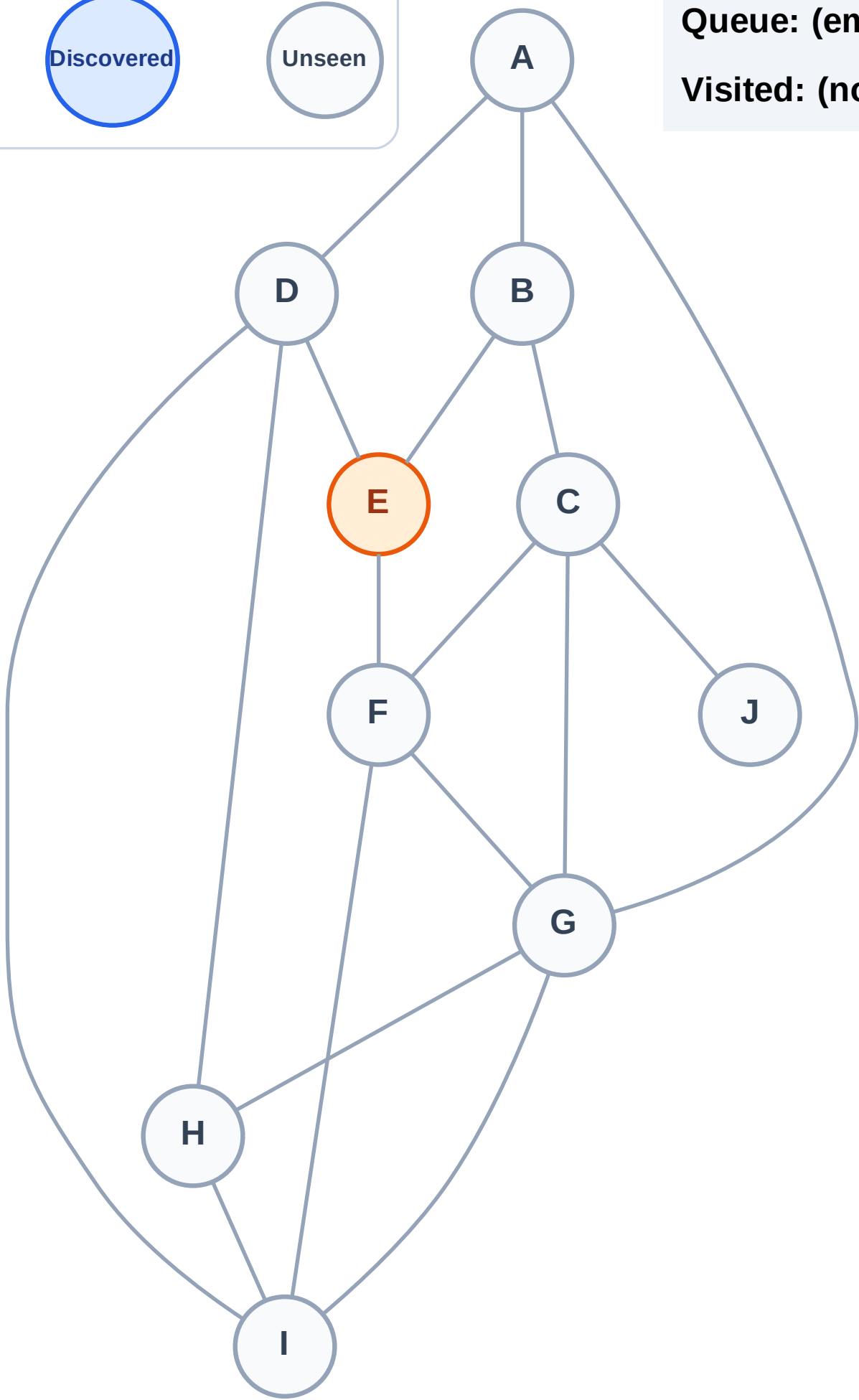
Step 2: Process node E

Legend



Queue: (empty)

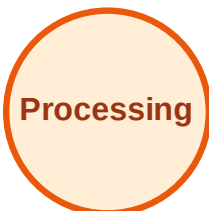
Visited: (none)



BFS Traversal

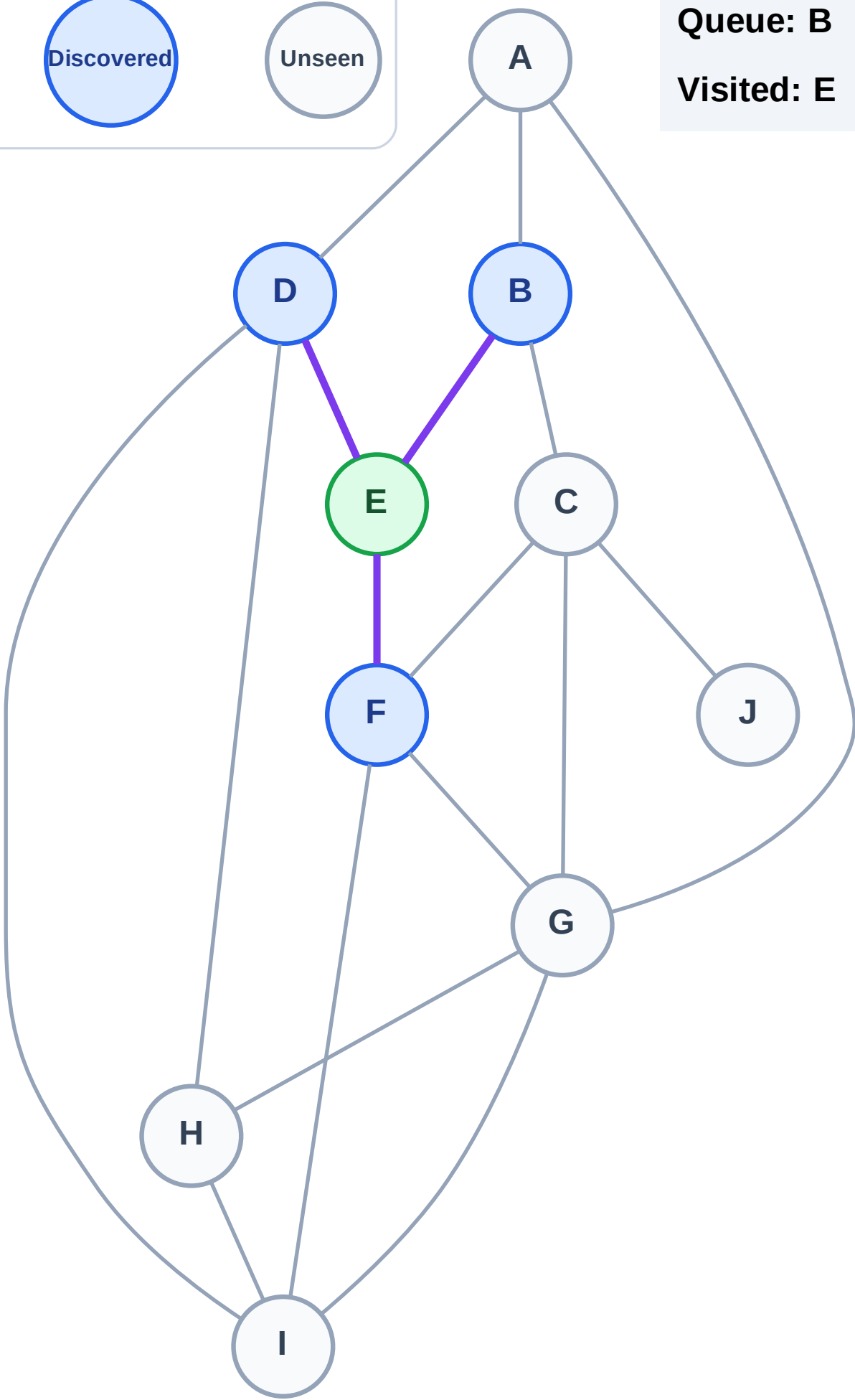
Step 3: Discover B, D, F from E

Legend



Queue: B → D → F

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

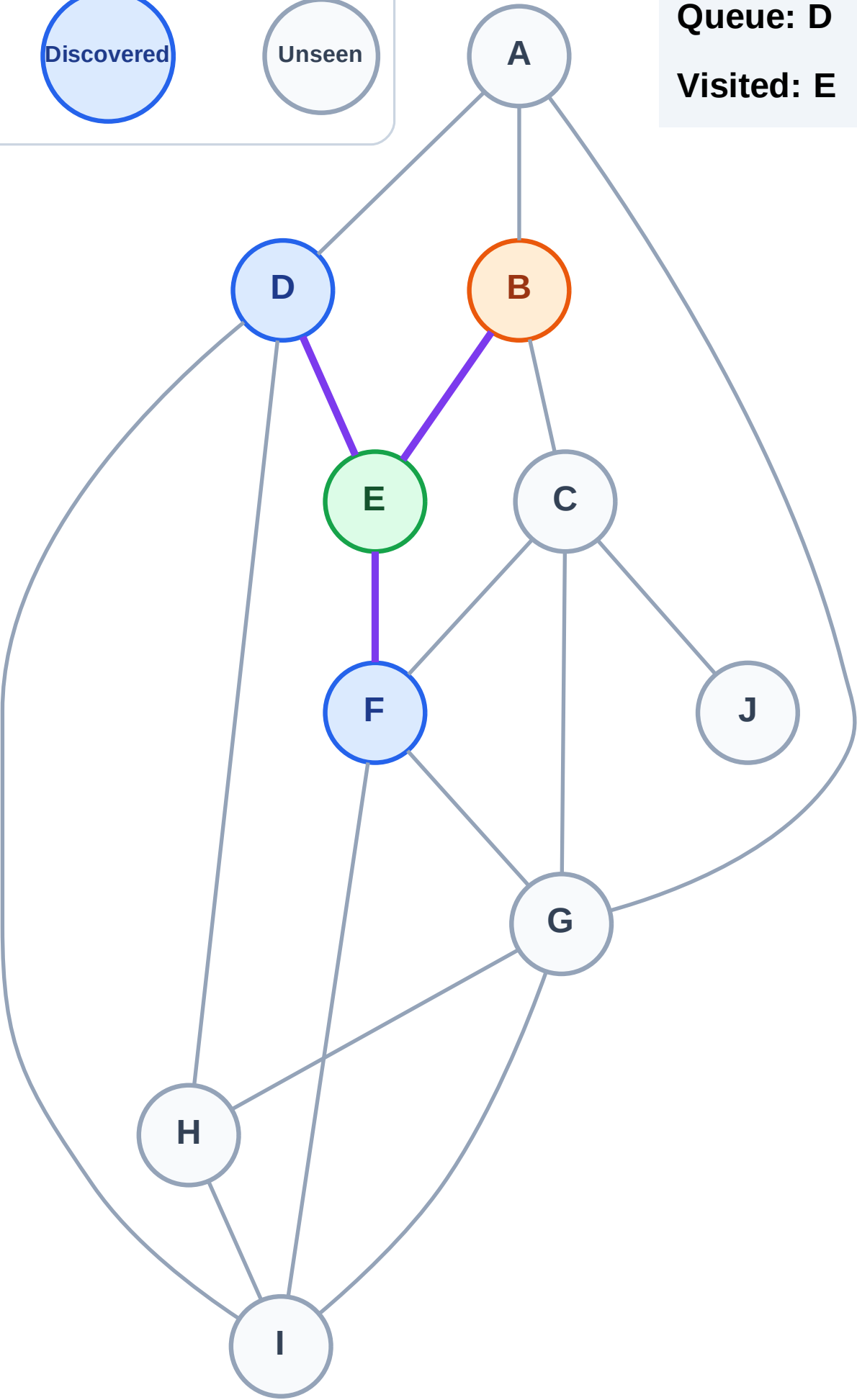
Processing

Discovered

Unseen

Queue: D → F

Visited: E



BFS Traversal

Step 5: Discover A, C from B

Legend

Complete

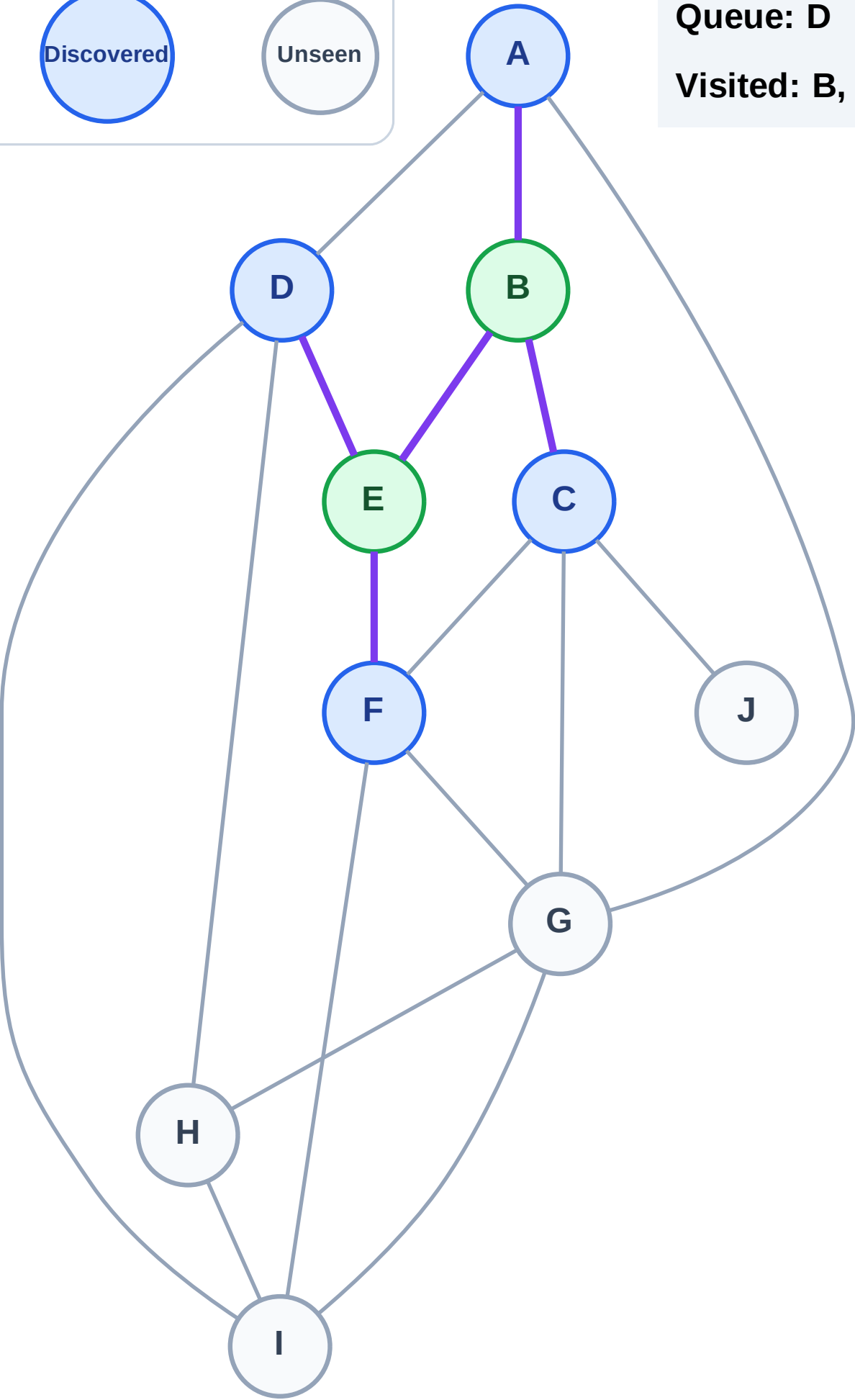
Processing

Discovered

Unseen

Queue: D → F → A → C

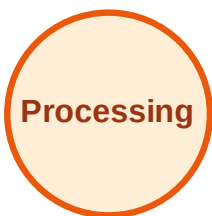
Visited: B, E



BFS Traversal

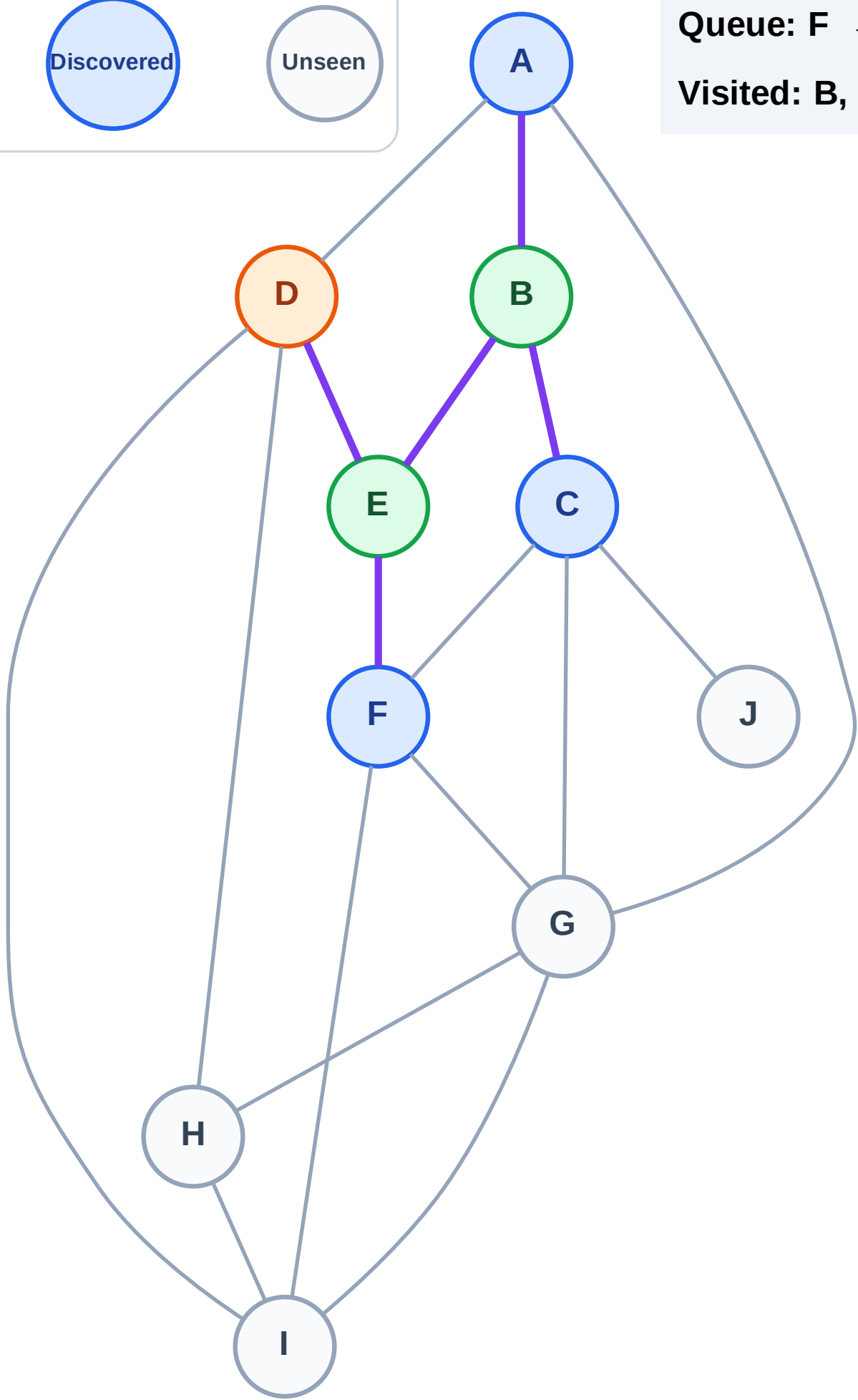
Step 6: Process node D

Legend



Queue: F → A → C

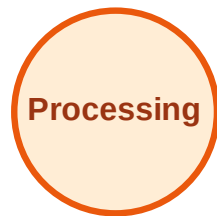
Visited: B, E



BFS Traversal

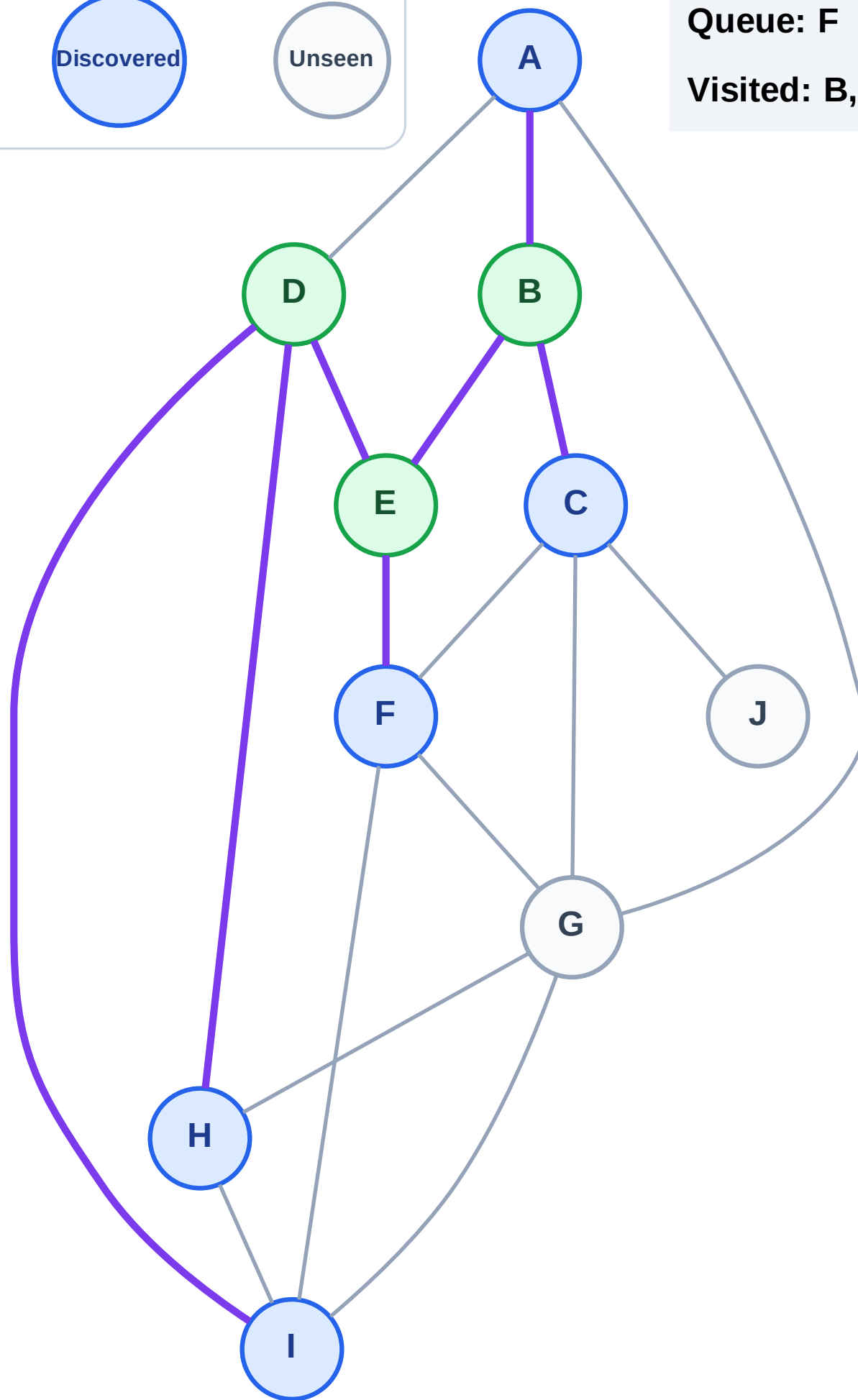
Step 7: Discover H, I from D

Legend



Queue: F → A → C → H → I

Visited: B, D, E



BFS Traversal

Step 8: Process node F

Legend

Complete

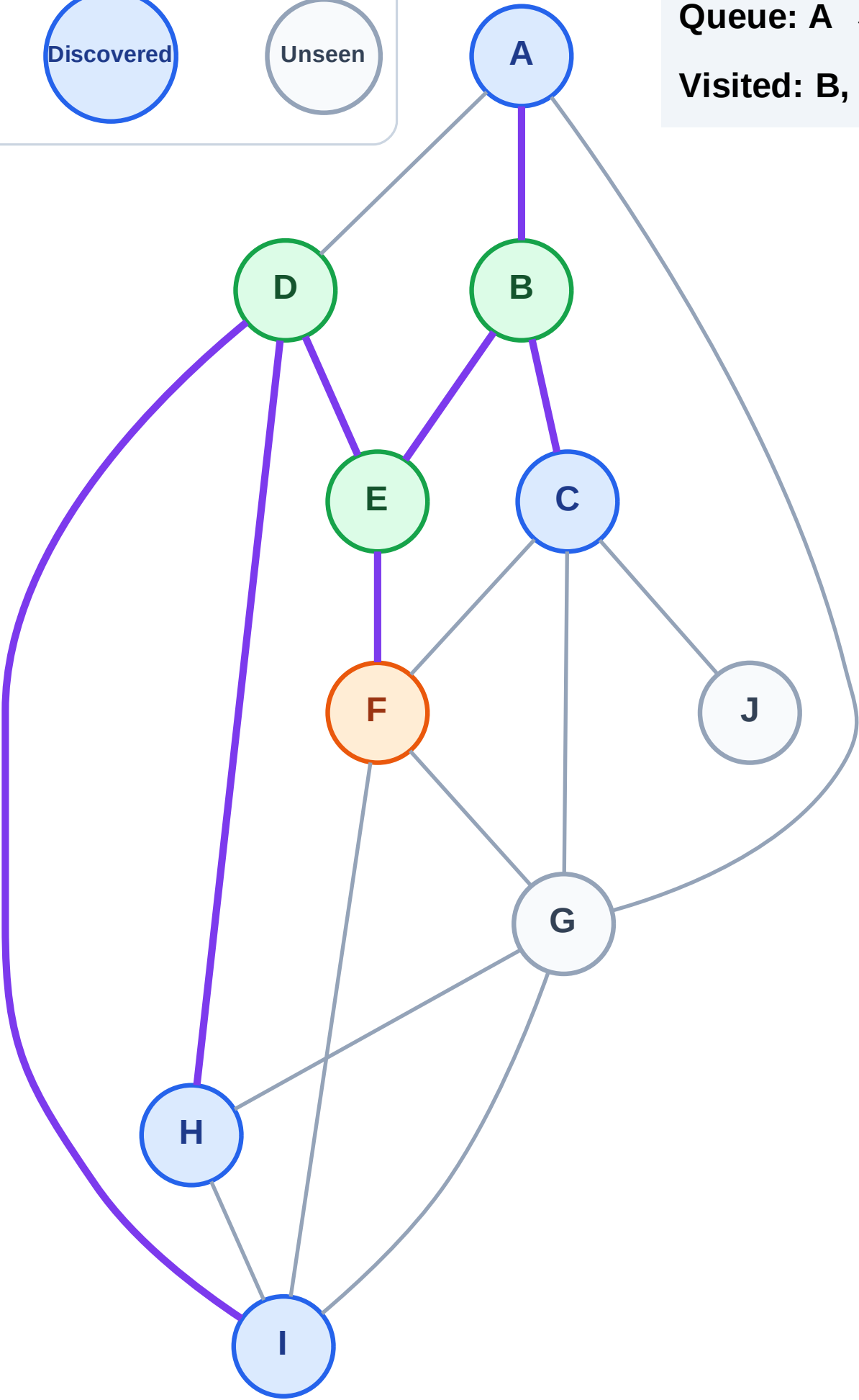
Processing

Discovered

Unseen

Queue: A → C → H → I

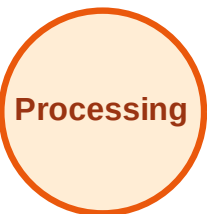
Visited: B, D, E



BFS Traversal

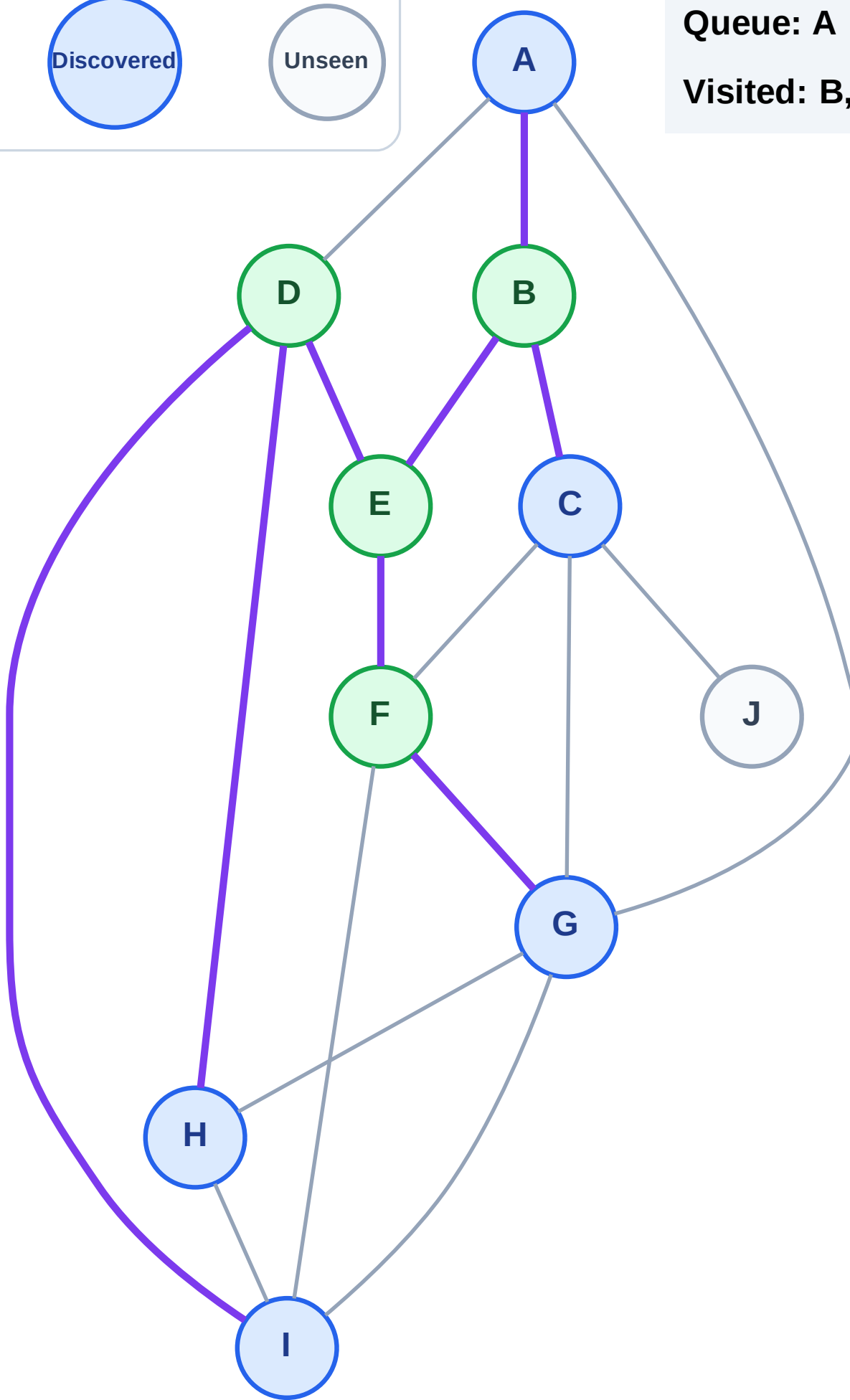
Step 9: Discover G from F

Legend



Queue: A → C → H → I → G

Visited: B, D, E, F



BFS Traversal

Step 10: Process node A

Legend

Complete

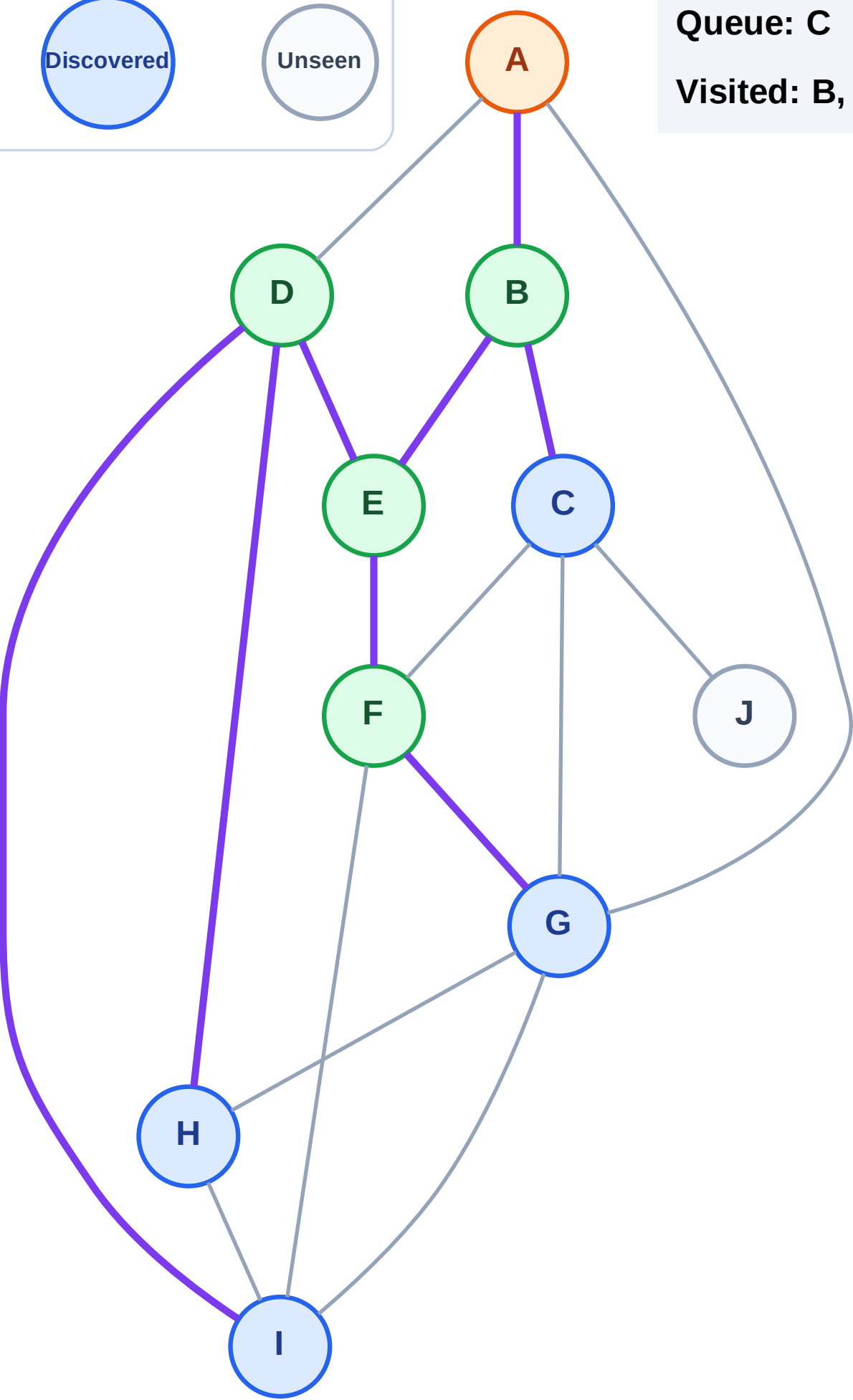
Processing

Discovered

Unseen

Queue: C → H → I → G

Visited: B, D, E, F



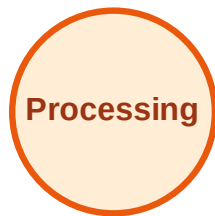
BFS Traversal

Step 11: No new neighbors from A

Legend



Complete



Processing

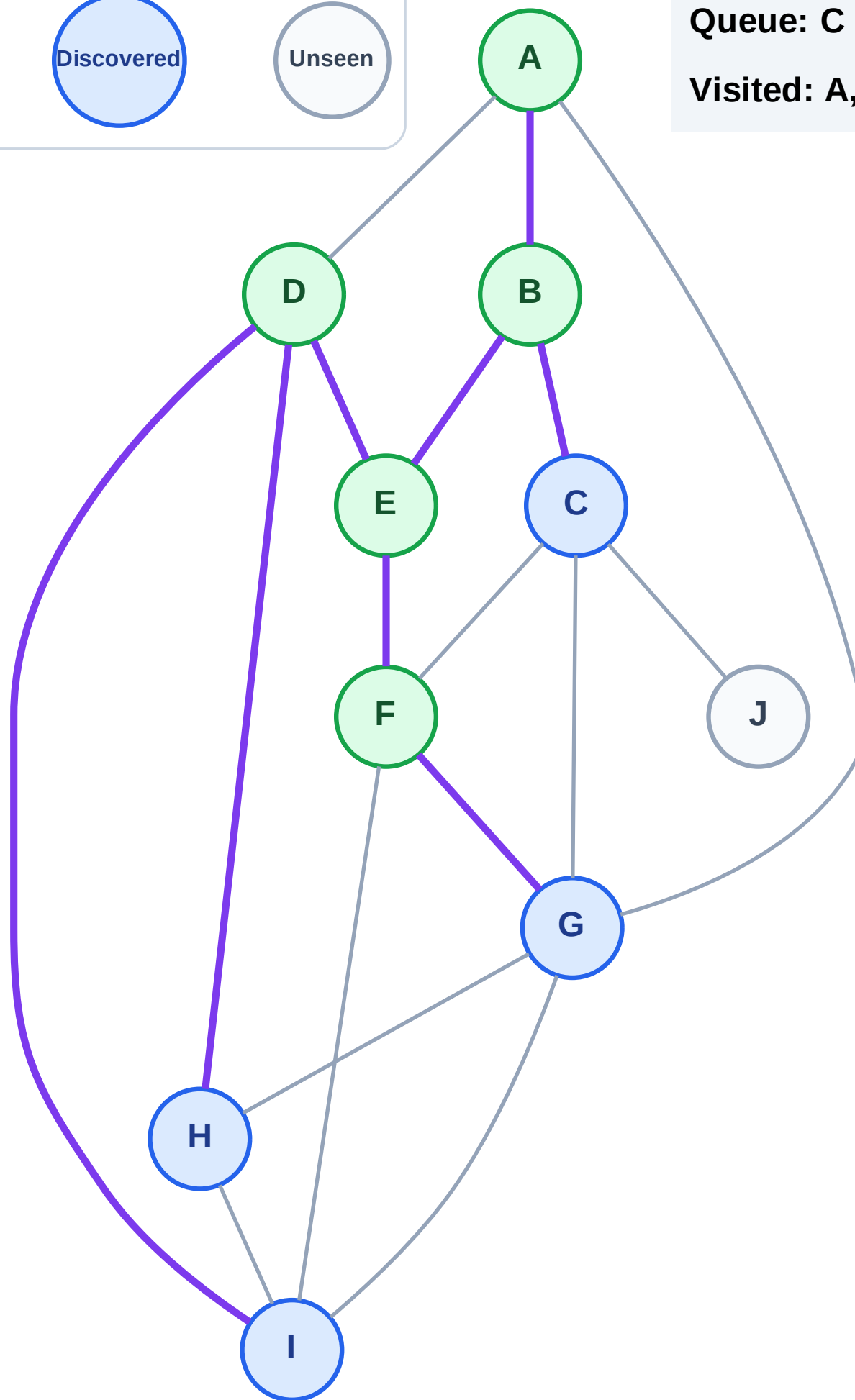


Discovered



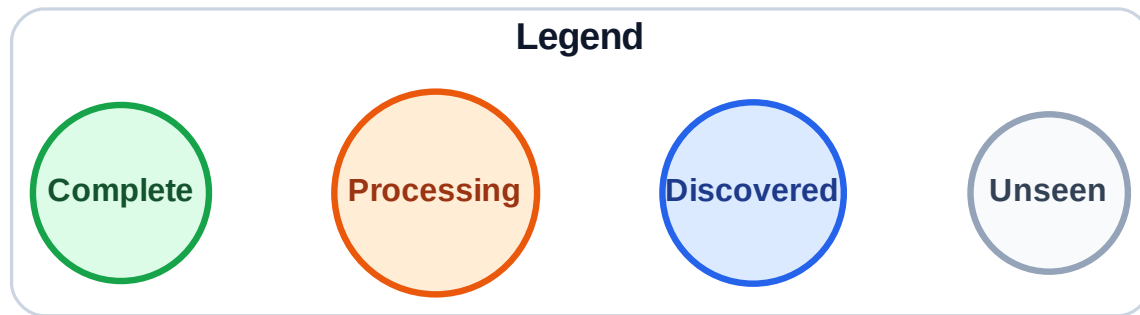
Unseen

Queue: C → H → I → G
Visited: A, B, D, E, F

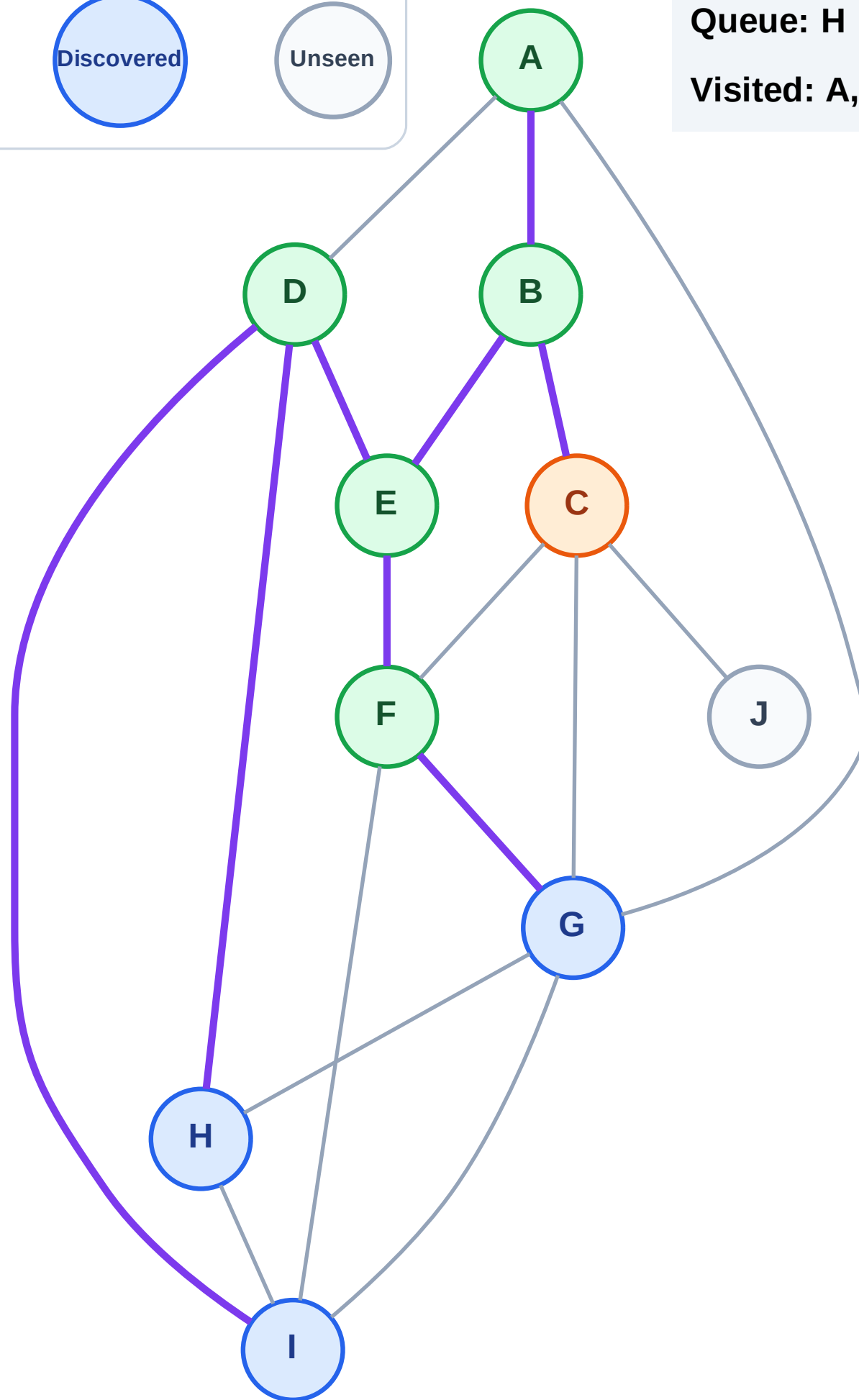


BFS Traversal

Step 12: Process node C



Queue: H → I → G
Visited: A, B, D, E, F



BFS Traversal

Step 13: Discover J from C

Legend

Complete

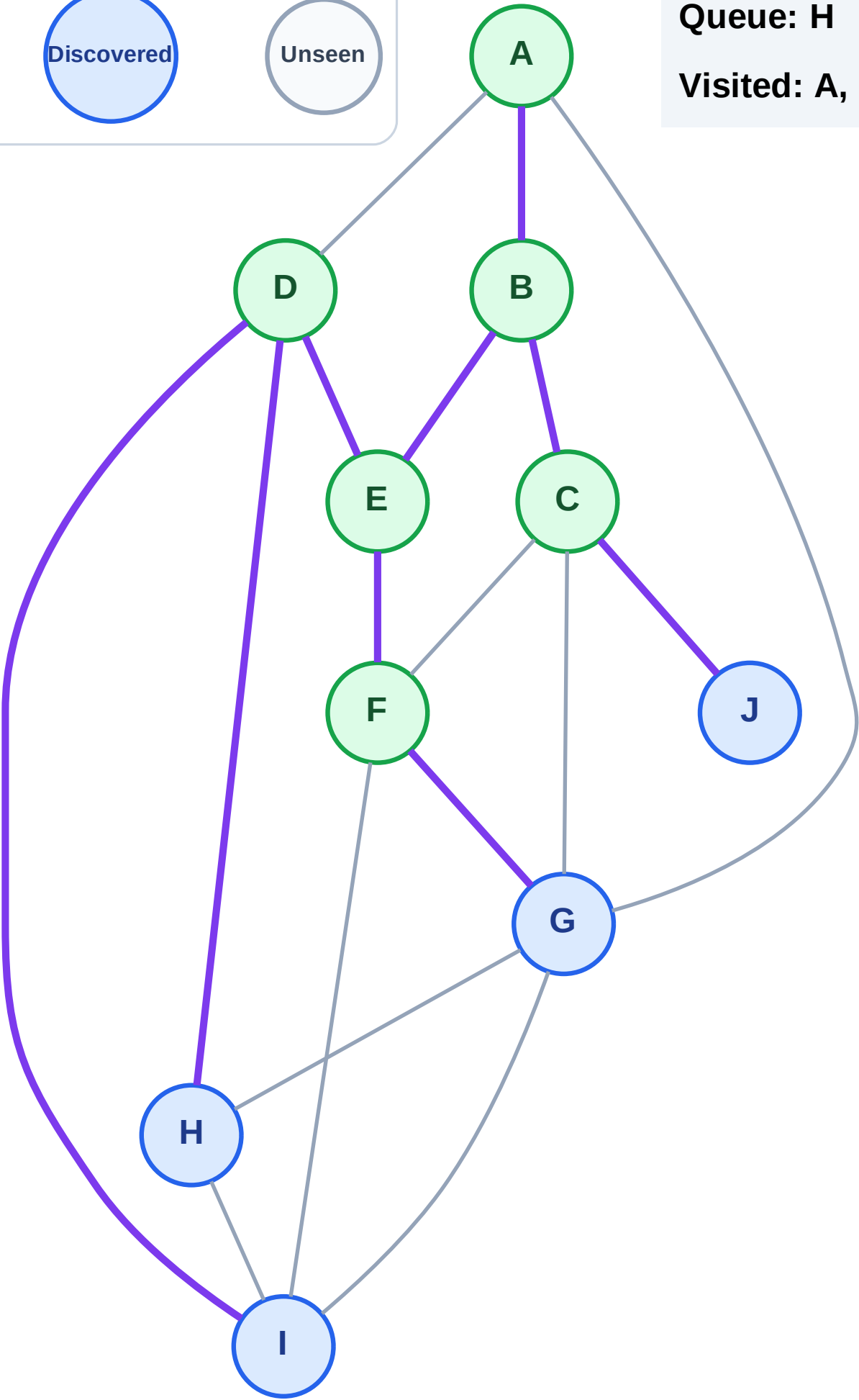
Processing

Discovered

Unseen

Queue: H → I → G → J

Visited: A, B, C, D, E, F



BFS Traversal

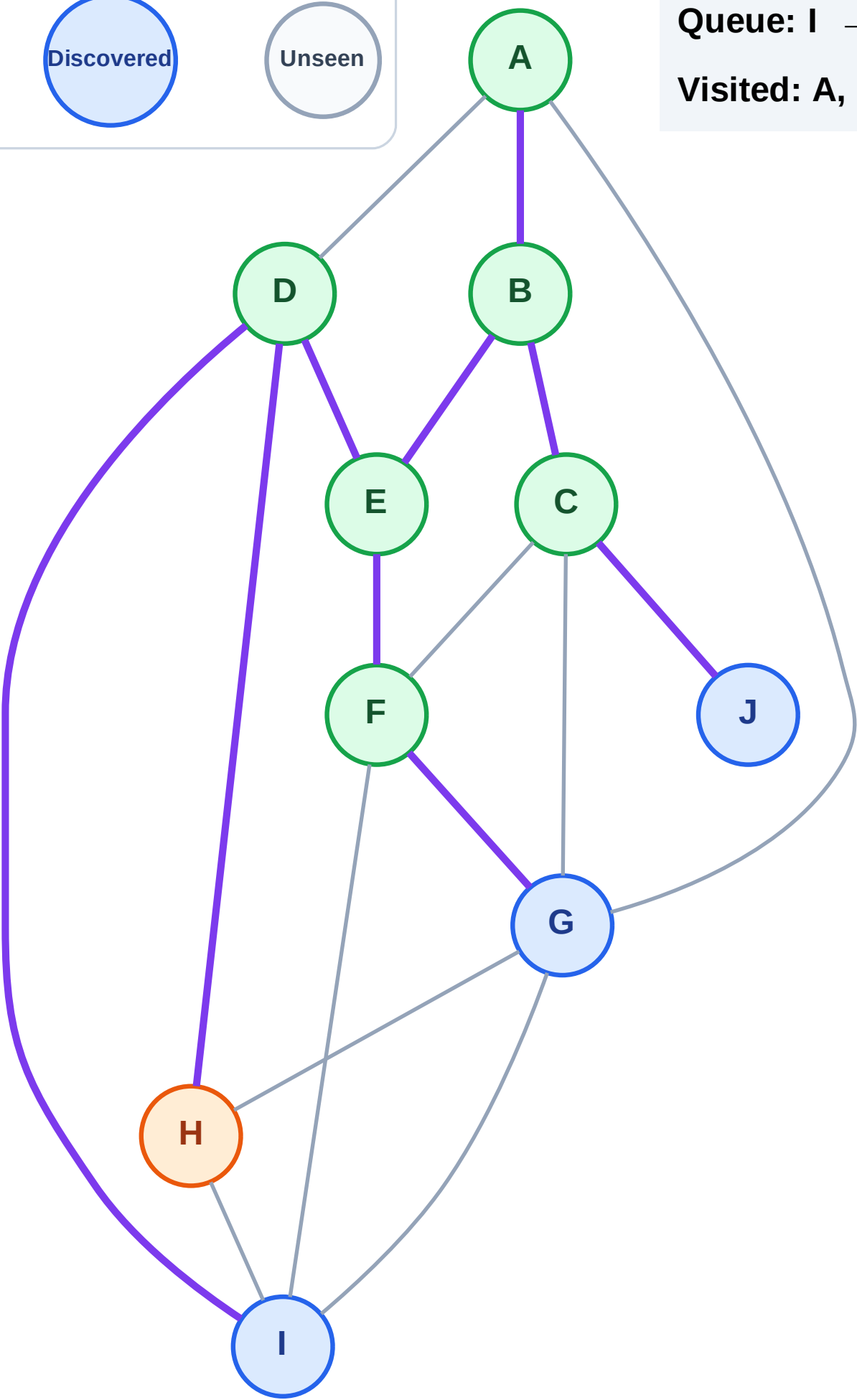
Step 14: Process node H

Legend



Queue: I → G → J

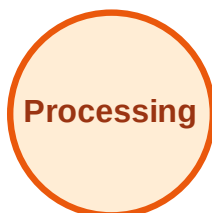
Visited: A, B, C, D, E, F



BFS Traversal

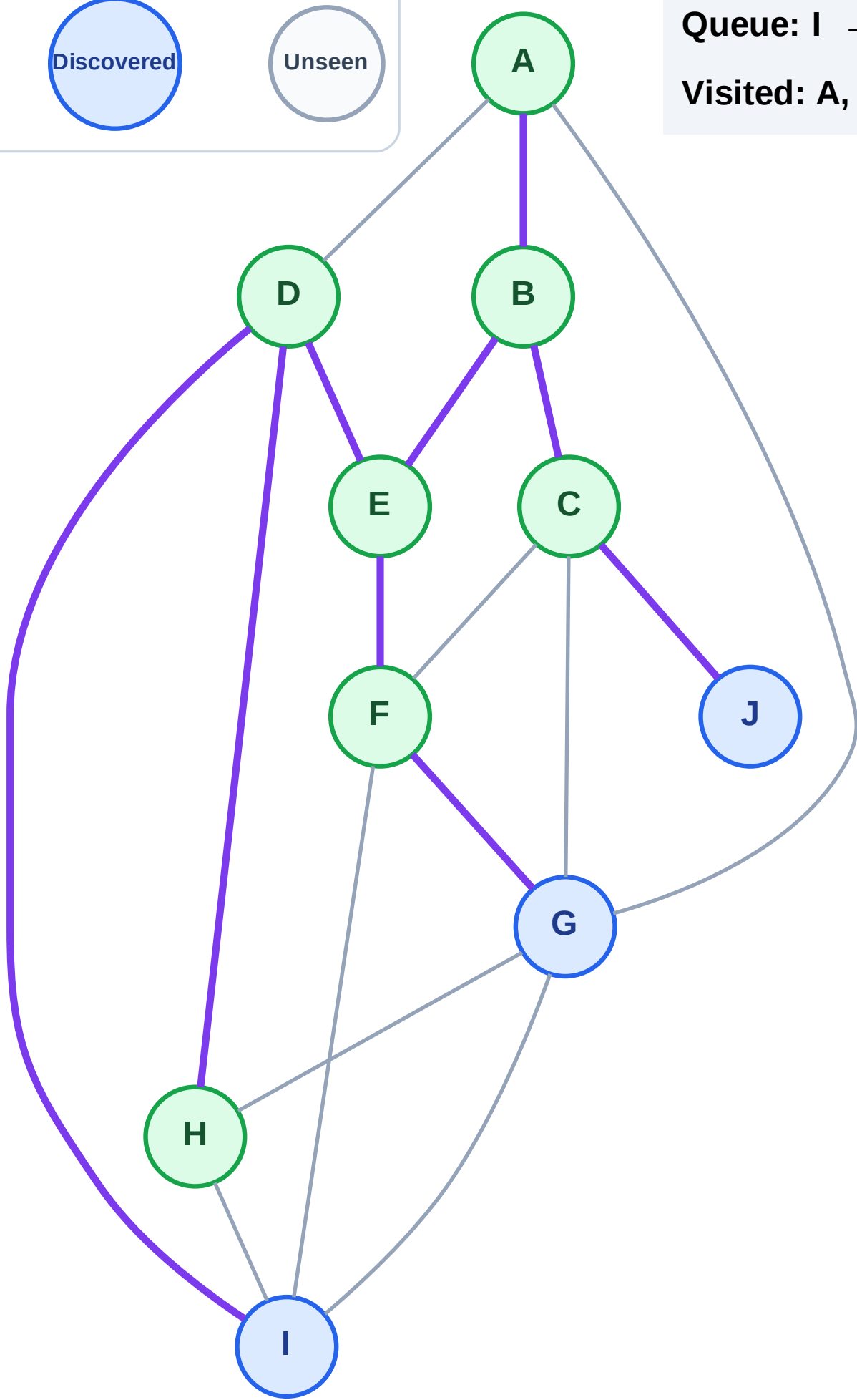
Step 15: No new neighbors from H

Legend



Queue: I → G → J

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 16: Process node I

Legend

Complete

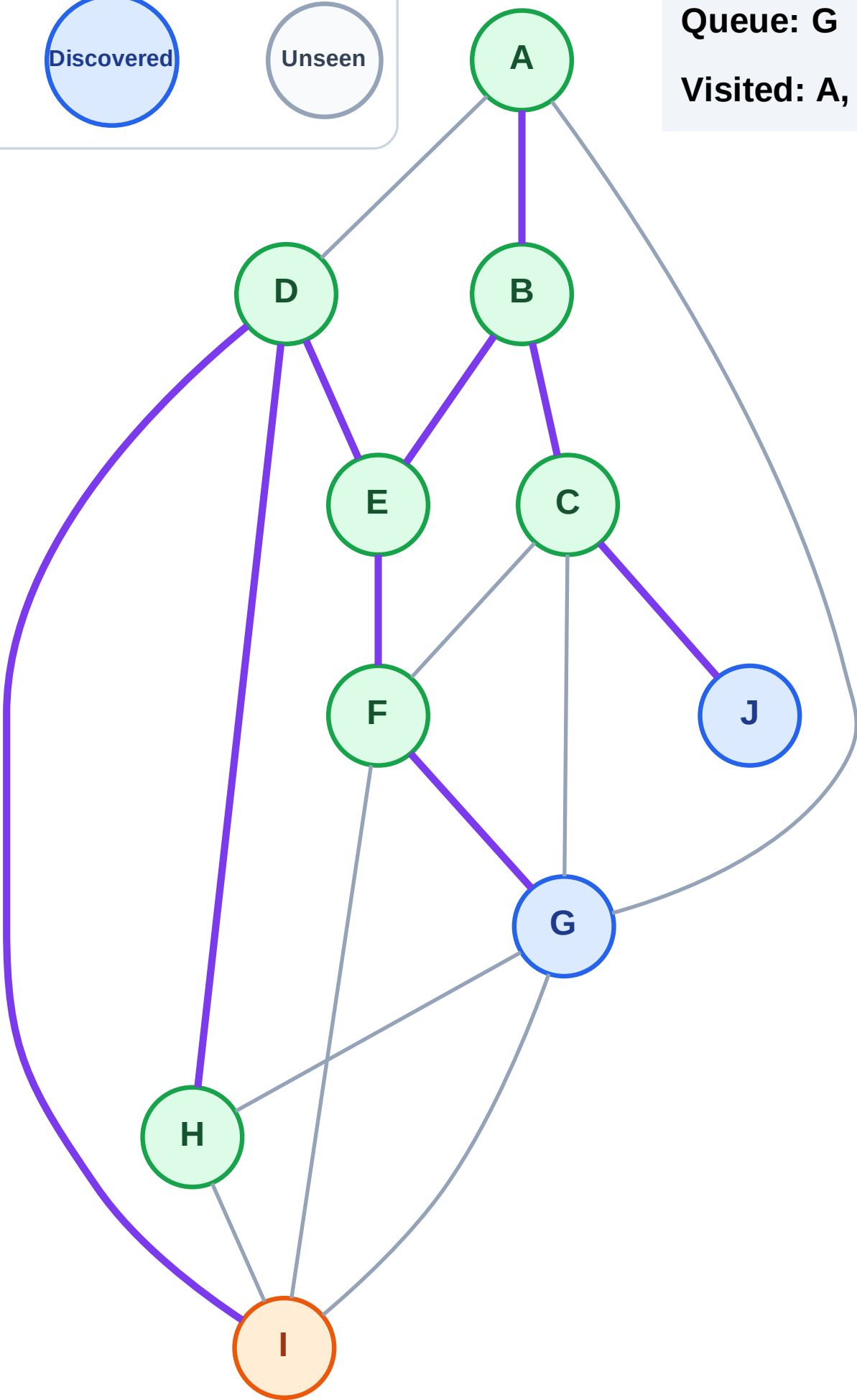
Processing

Discovered

Unseen

Queue: G → J

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 17: No new neighbors from I

Legend

Complete

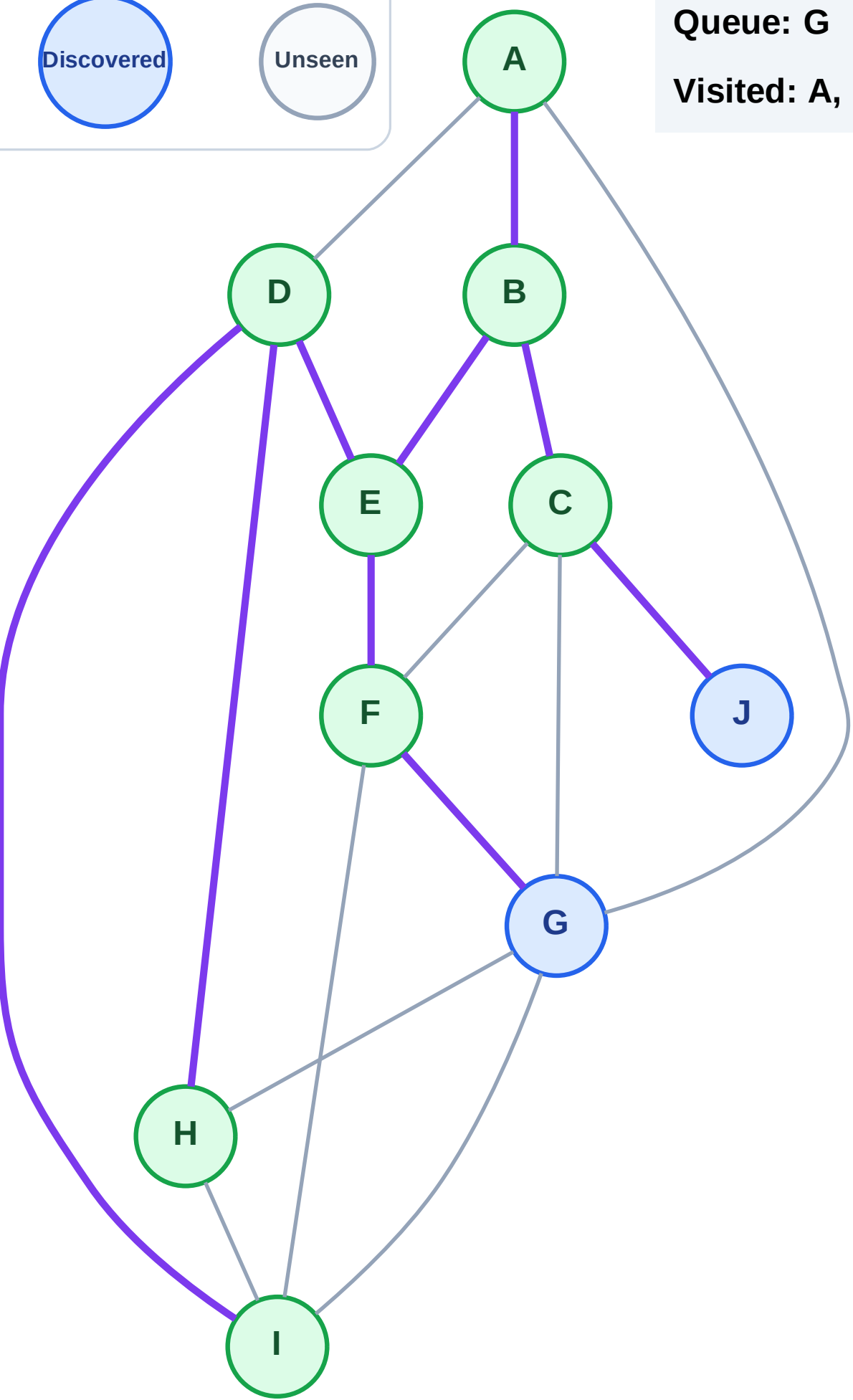
Processing

Discovered

Unseen

Queue: G → J

Visited: A, B, C, D, E, F, H, I



BFS Traversal

Step 18: Process node G

Legend

Complete

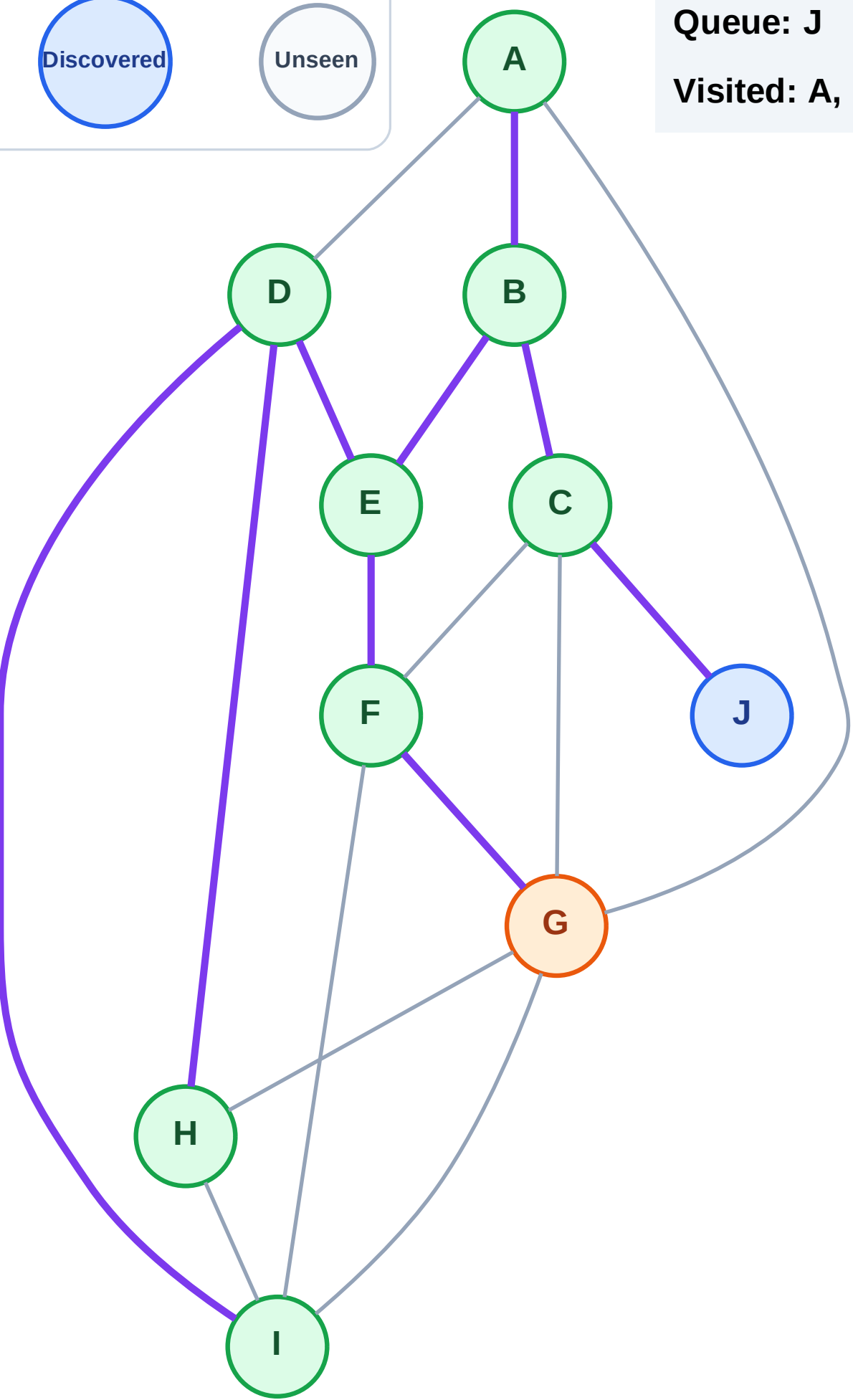
Processing

Discovered

Unseen

Queue: J

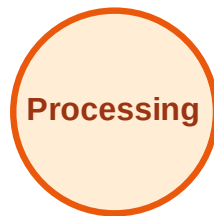
Visited: A, B, C, D, E, F, H, I



BFS Traversal

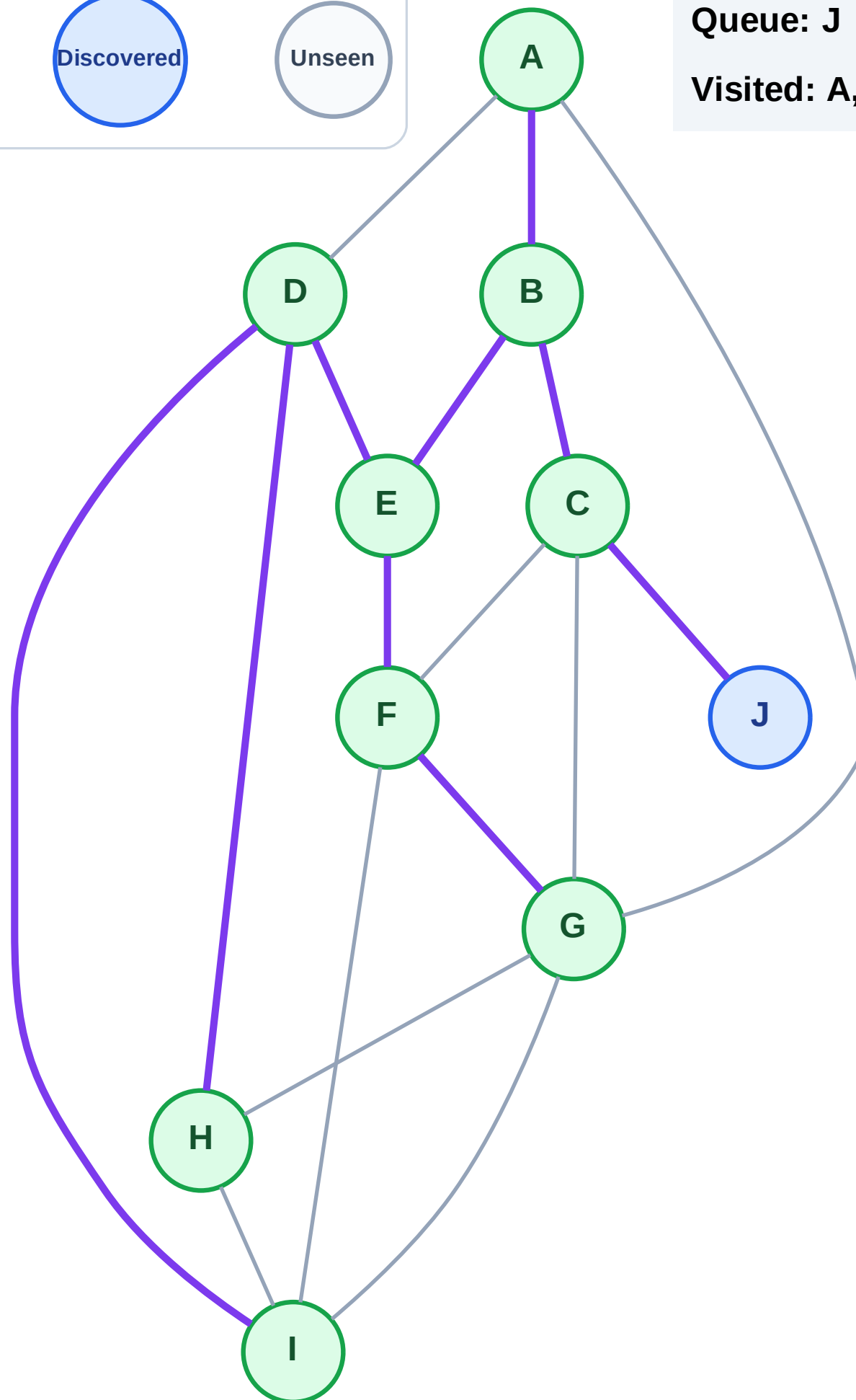
Step 19: No new neighbors from G

Legend



Queue: J

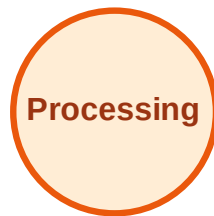
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

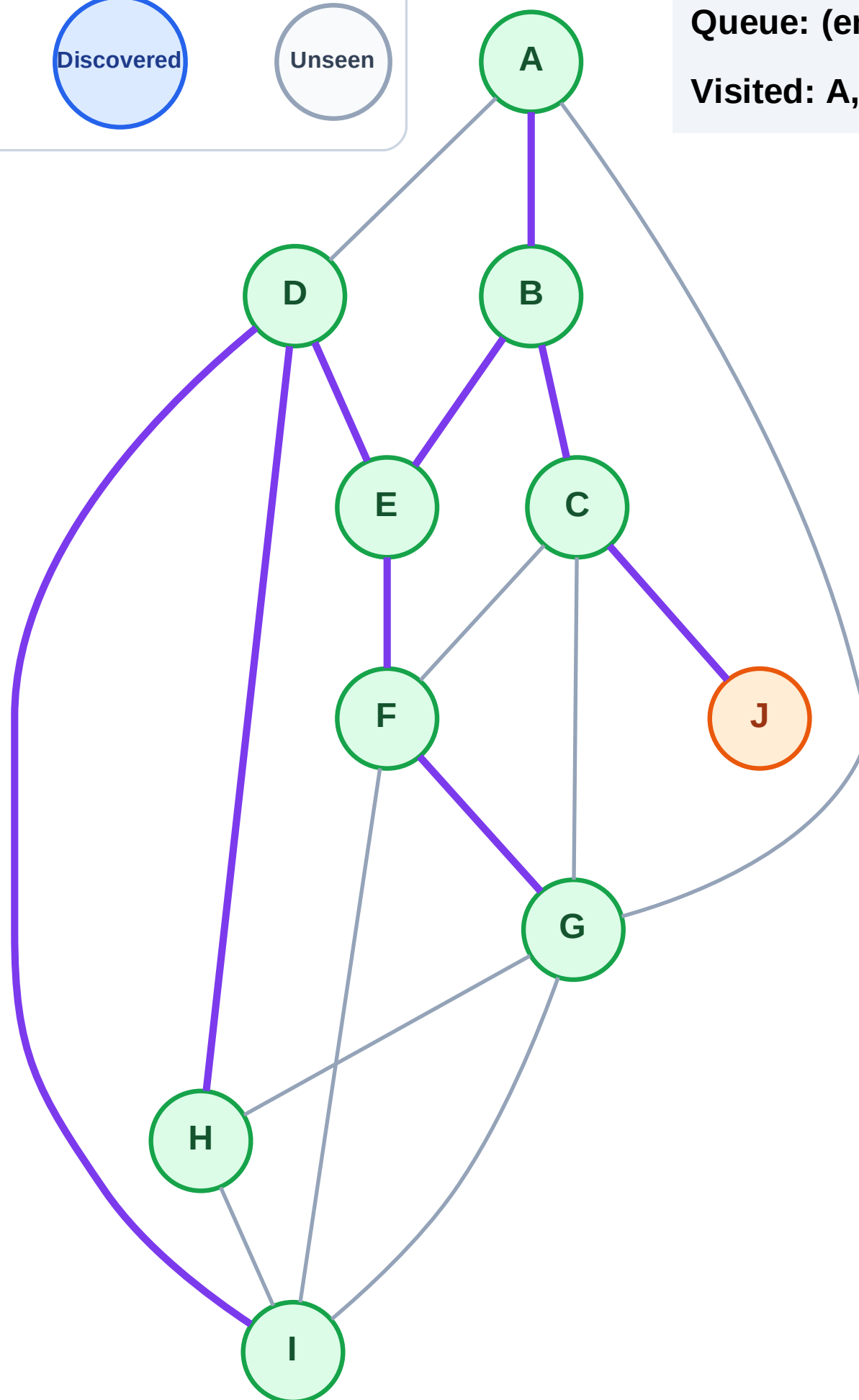
Step 20: Process node J

Legend



Queue: (empty)

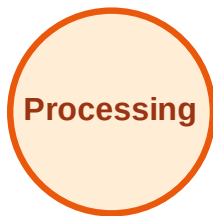
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

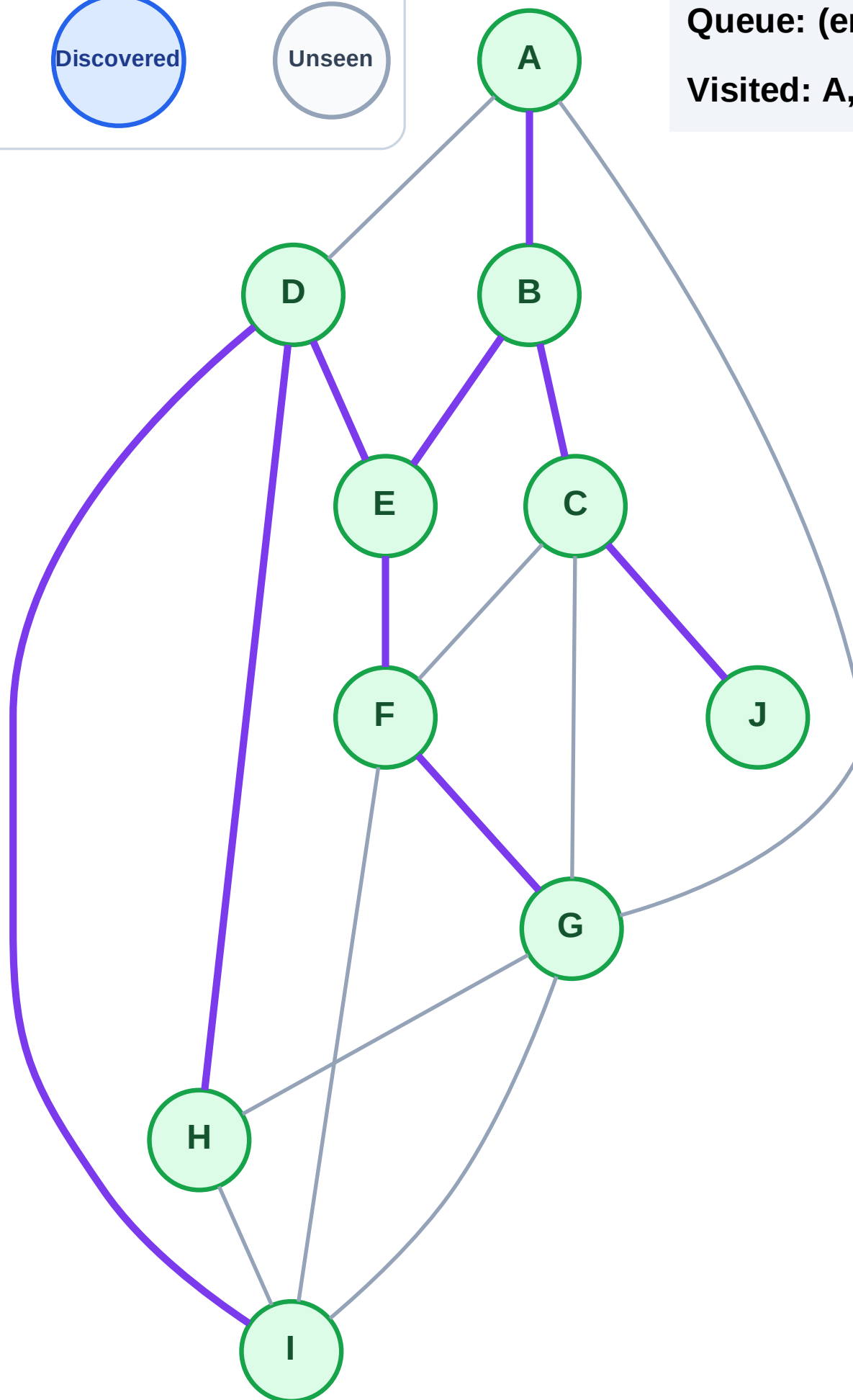
Step 21: No new neighbors from J

Legend



Queue: (empty)

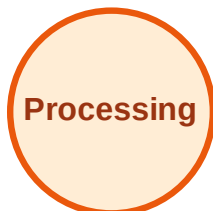
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

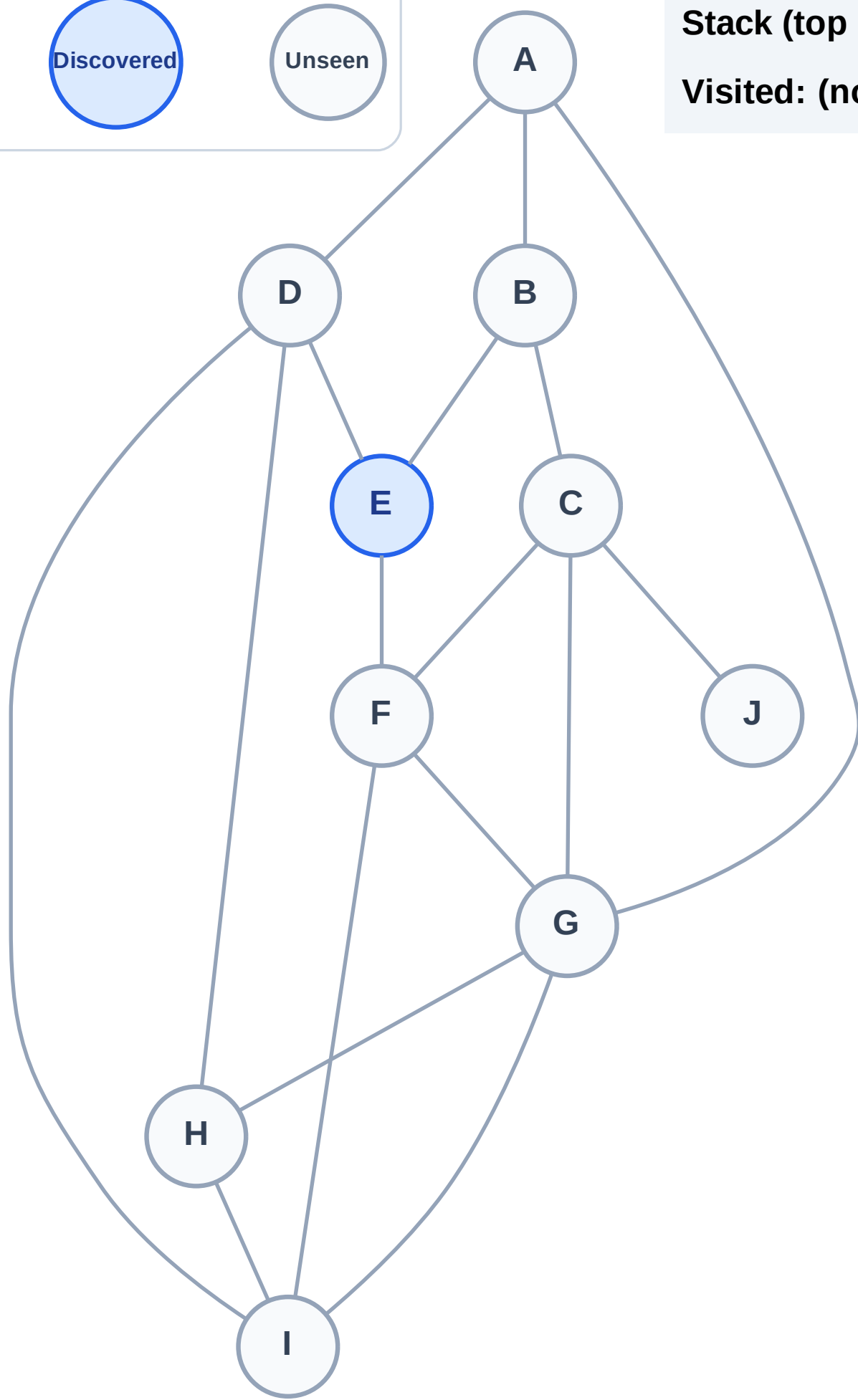
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

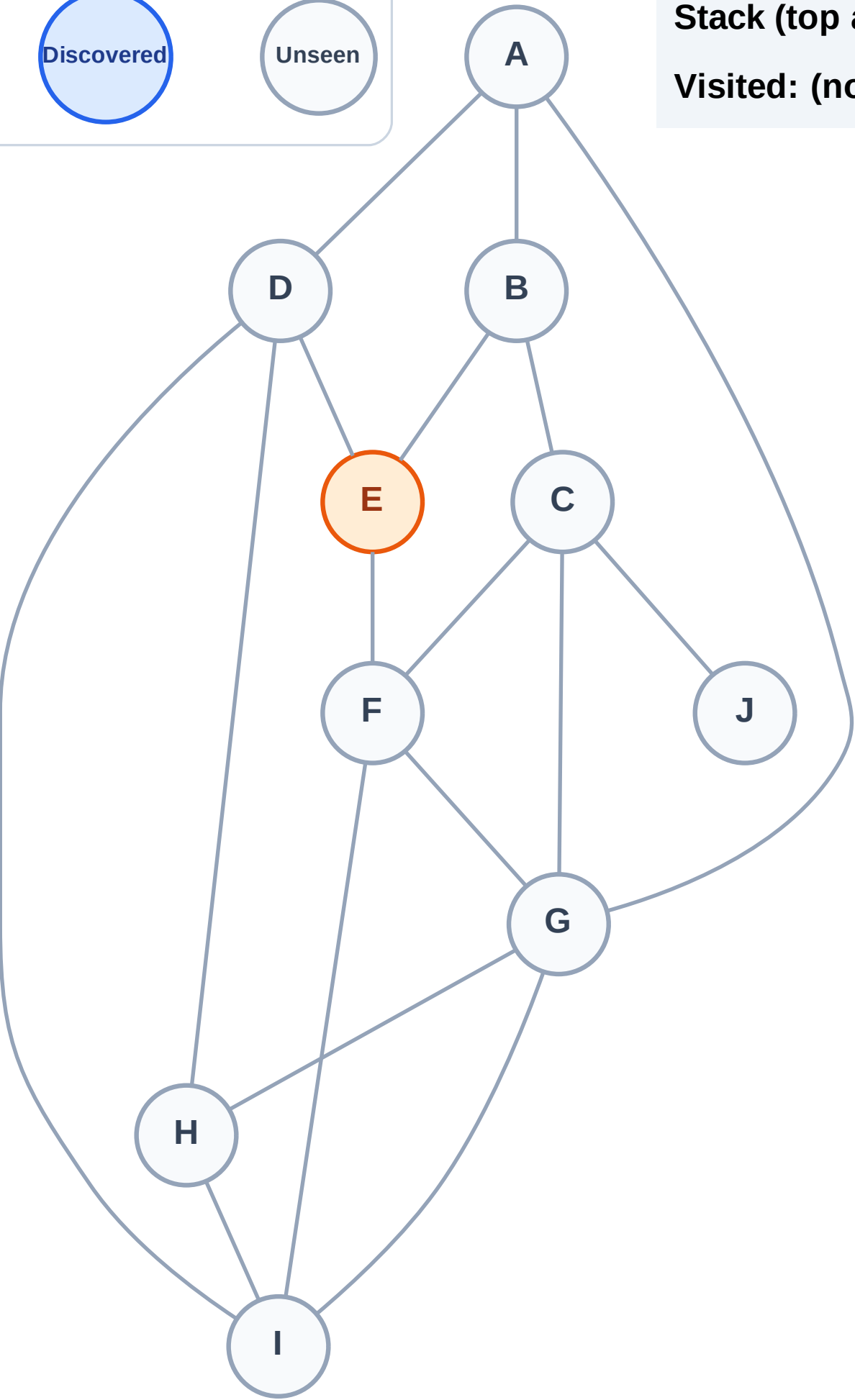
Complete

Processing

Discovered

Unseen

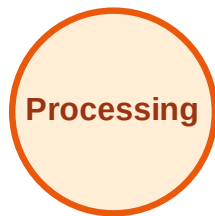
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

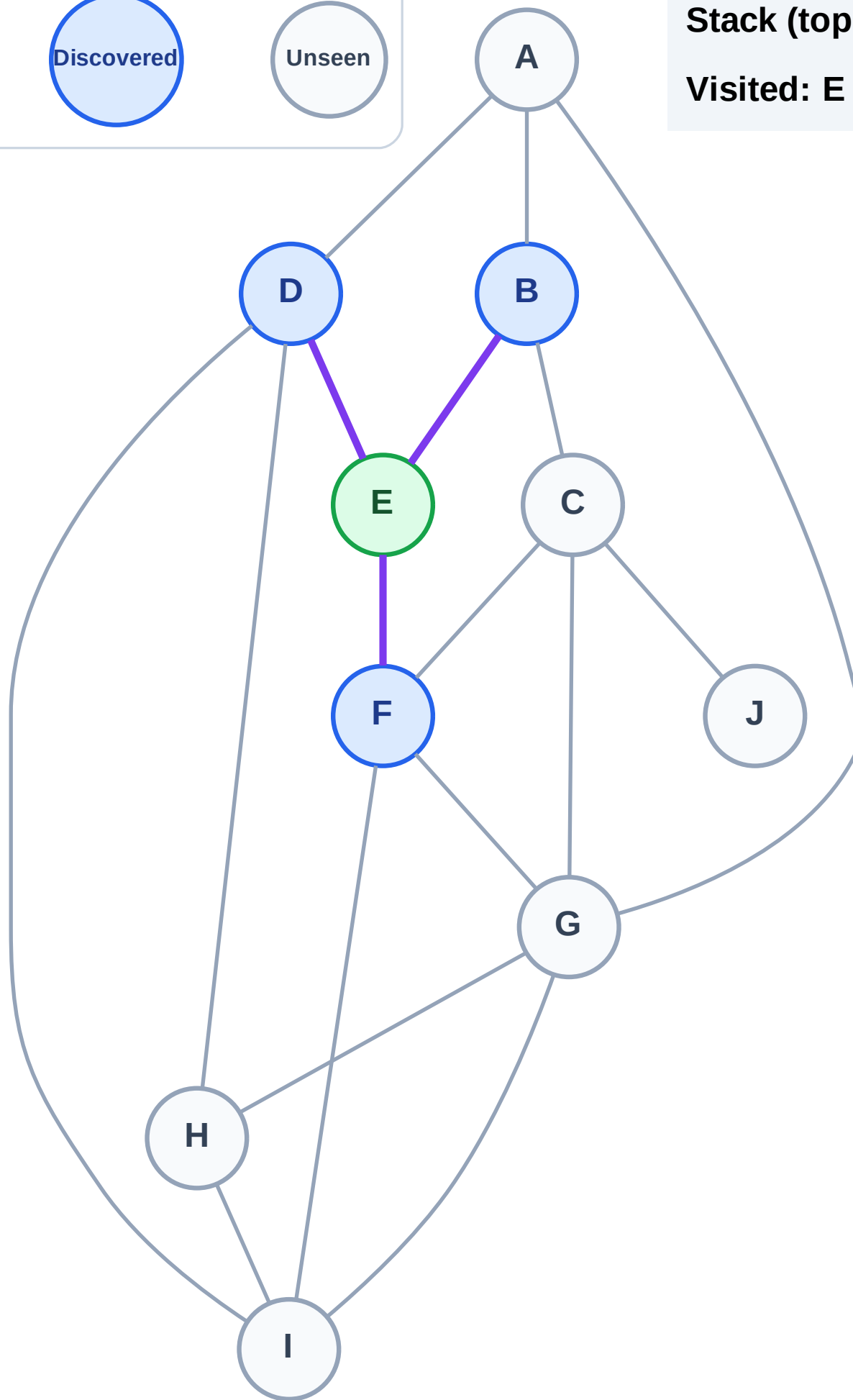
Step 3: Discover F, D, B from E

Legend



Stack (top at right): F | D | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

Complete

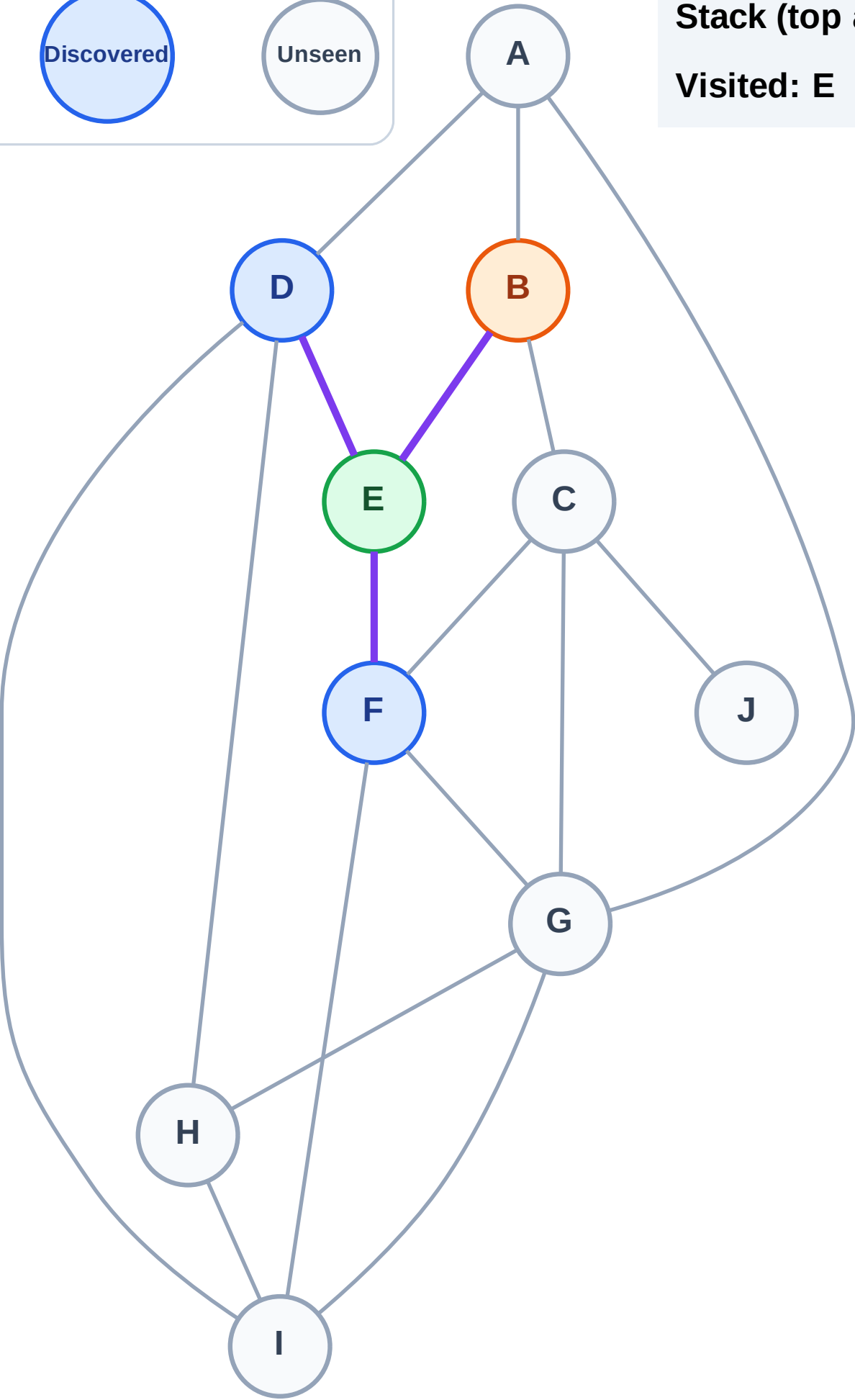
Processing

Discovered

Unseen

Stack (top at right): F | D

Visited: E



DFS Traversal

Step 5: Discover C, A from B

Legend

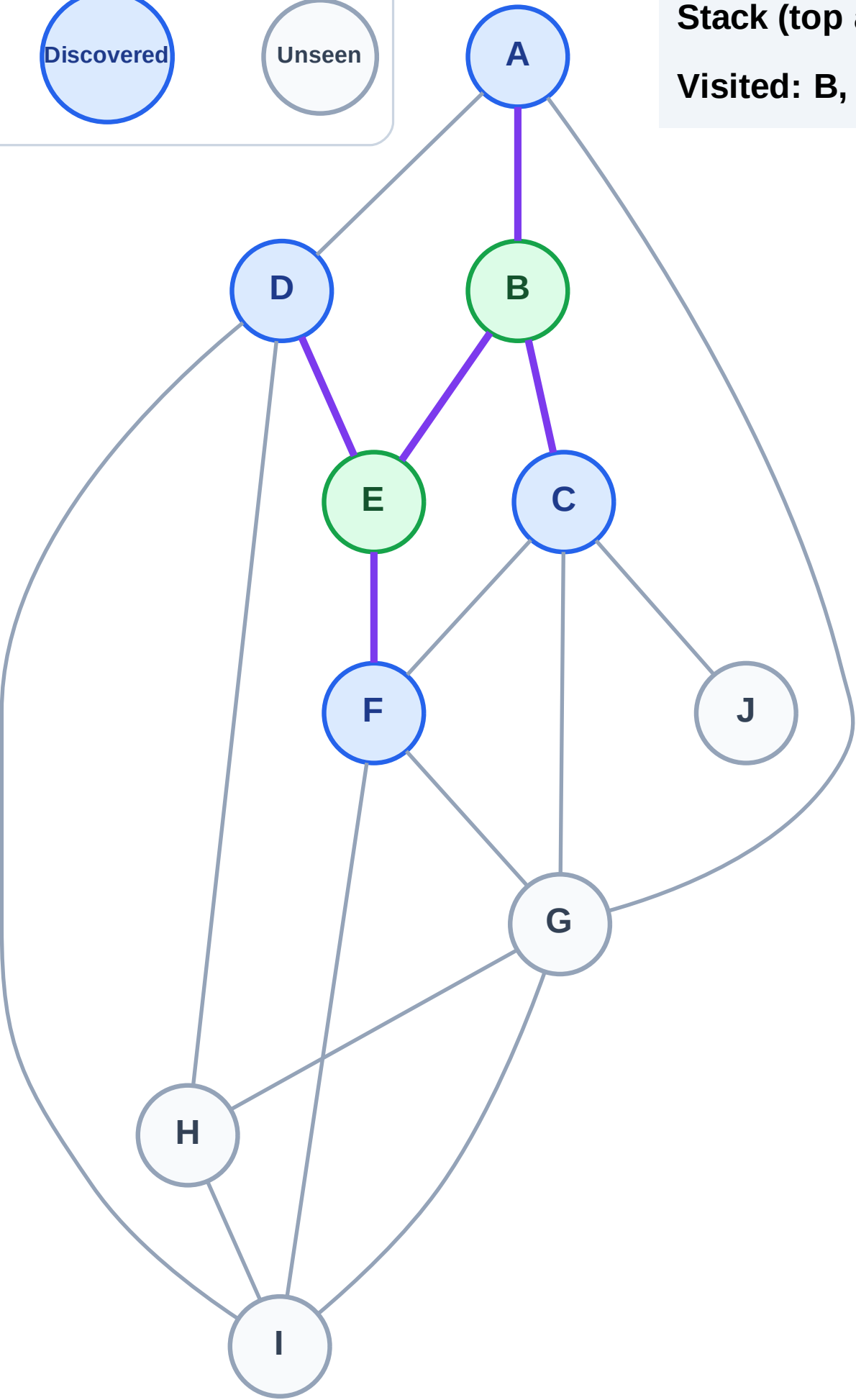
Complete

Processing

Discovered

Unseen

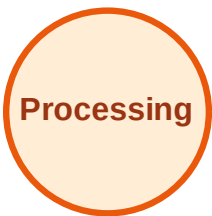
Stack (top at right): F | D | C | A
Visited: B, E



DFS Traversal

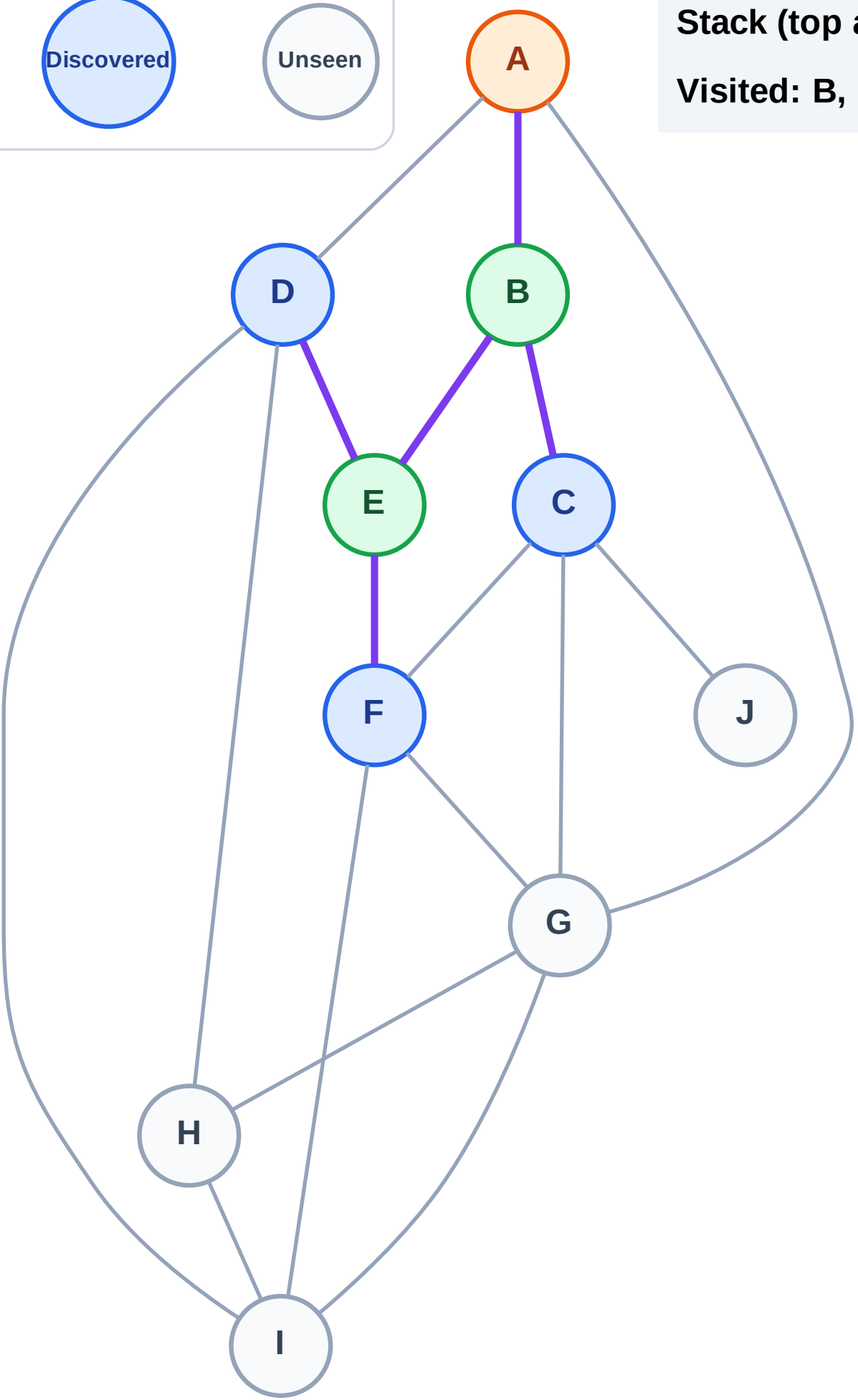
Step 6: Process node A

Legend



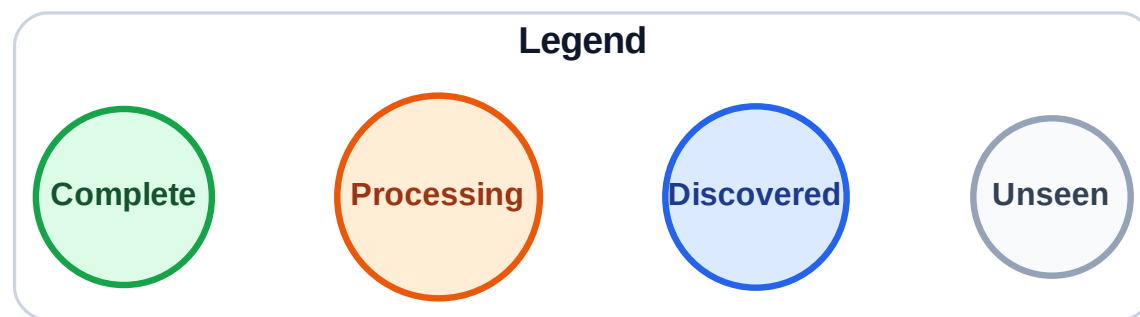
Stack (top at right): F | D | C

Visited: B, E

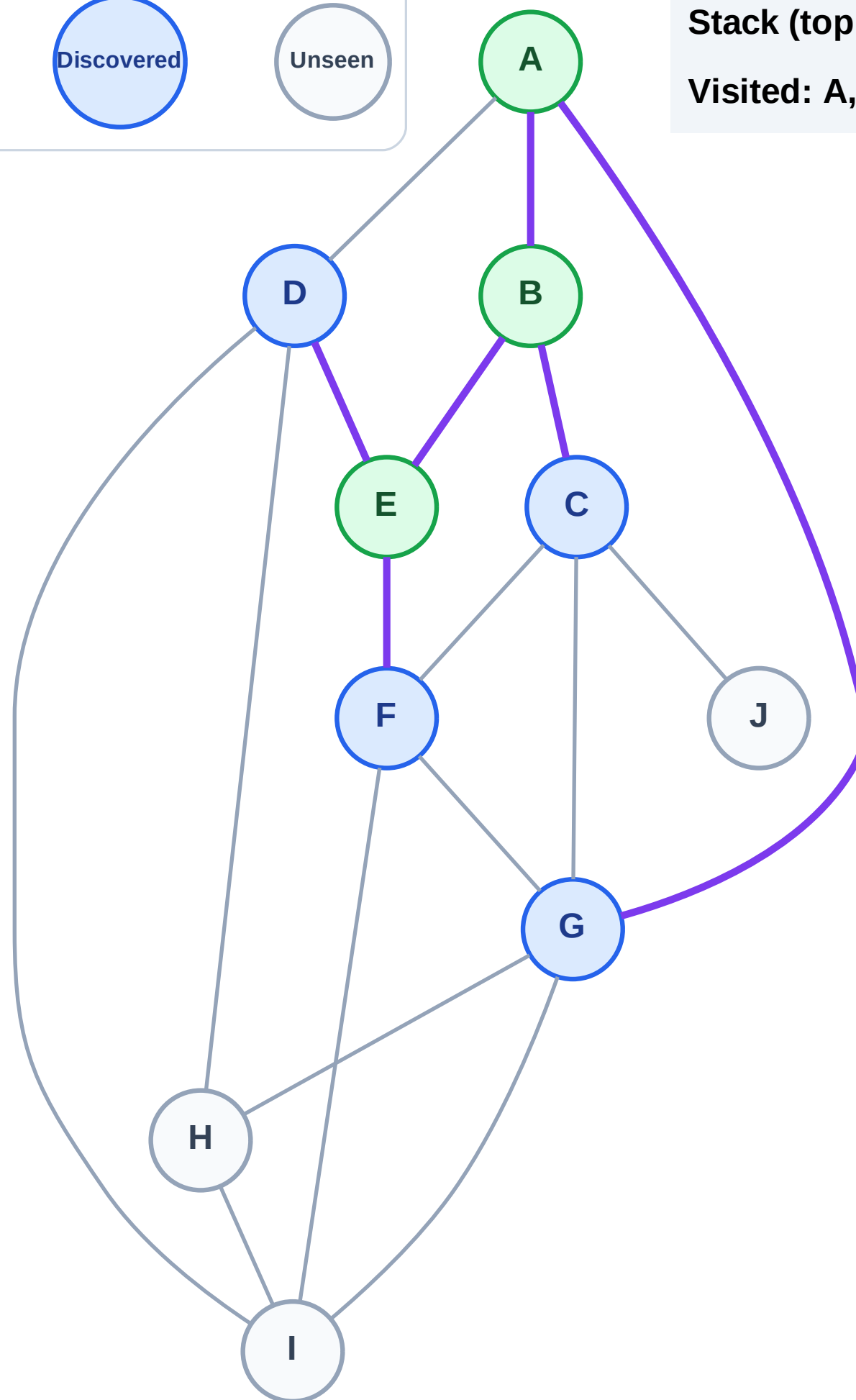


DFS Traversal

Step 7: Discover G from A



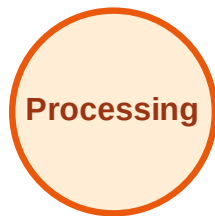
Stack (top at right): F | D | C | G
Visited: A, B, E



DFS Traversal

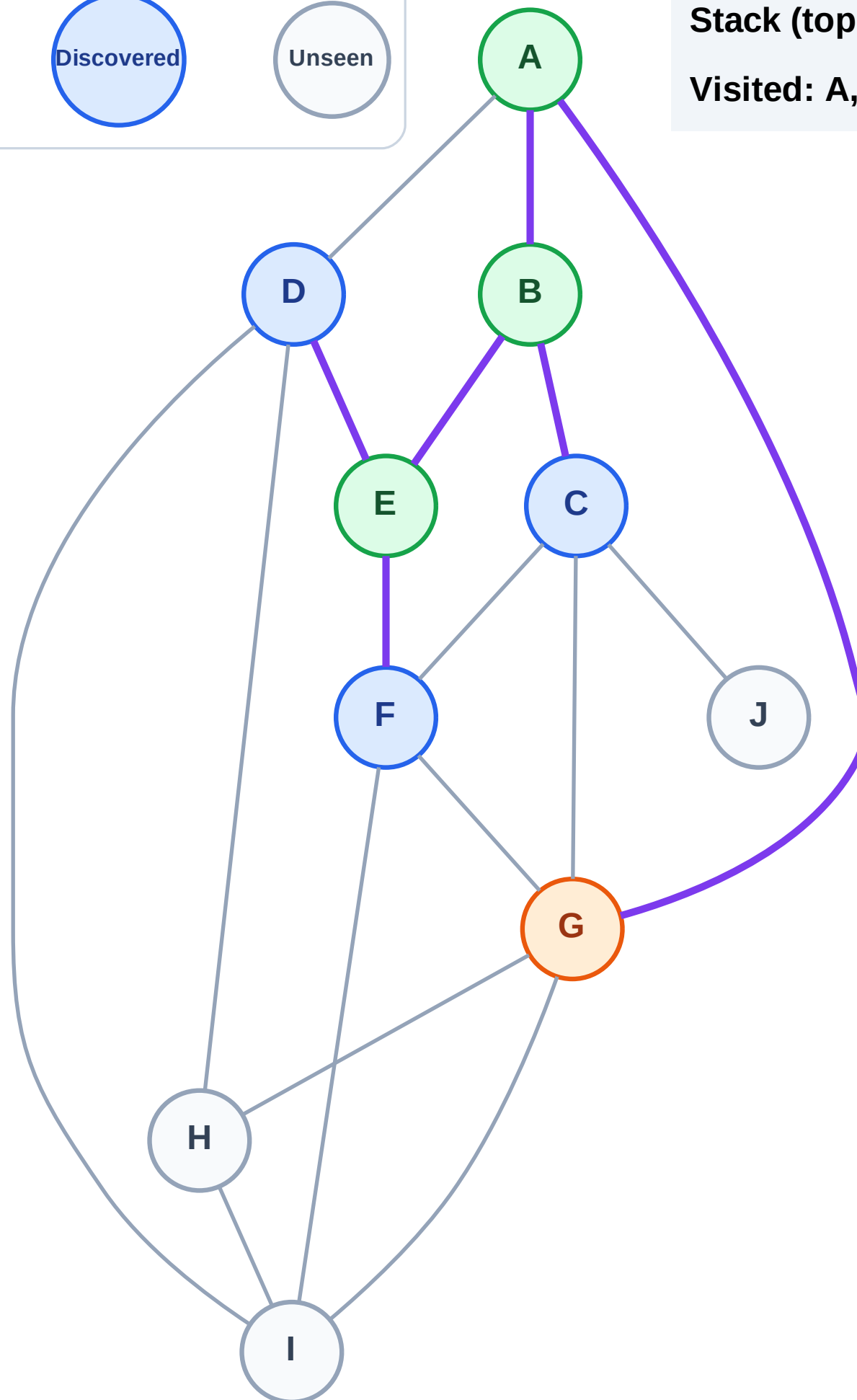
Step 8: Process node G

Legend



Stack (top at right): F | D | C

Visited: A, B, E



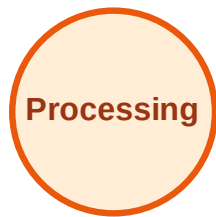
DFS Traversal

Step 9: Discover I, H from G

Legend



Complete



Processing



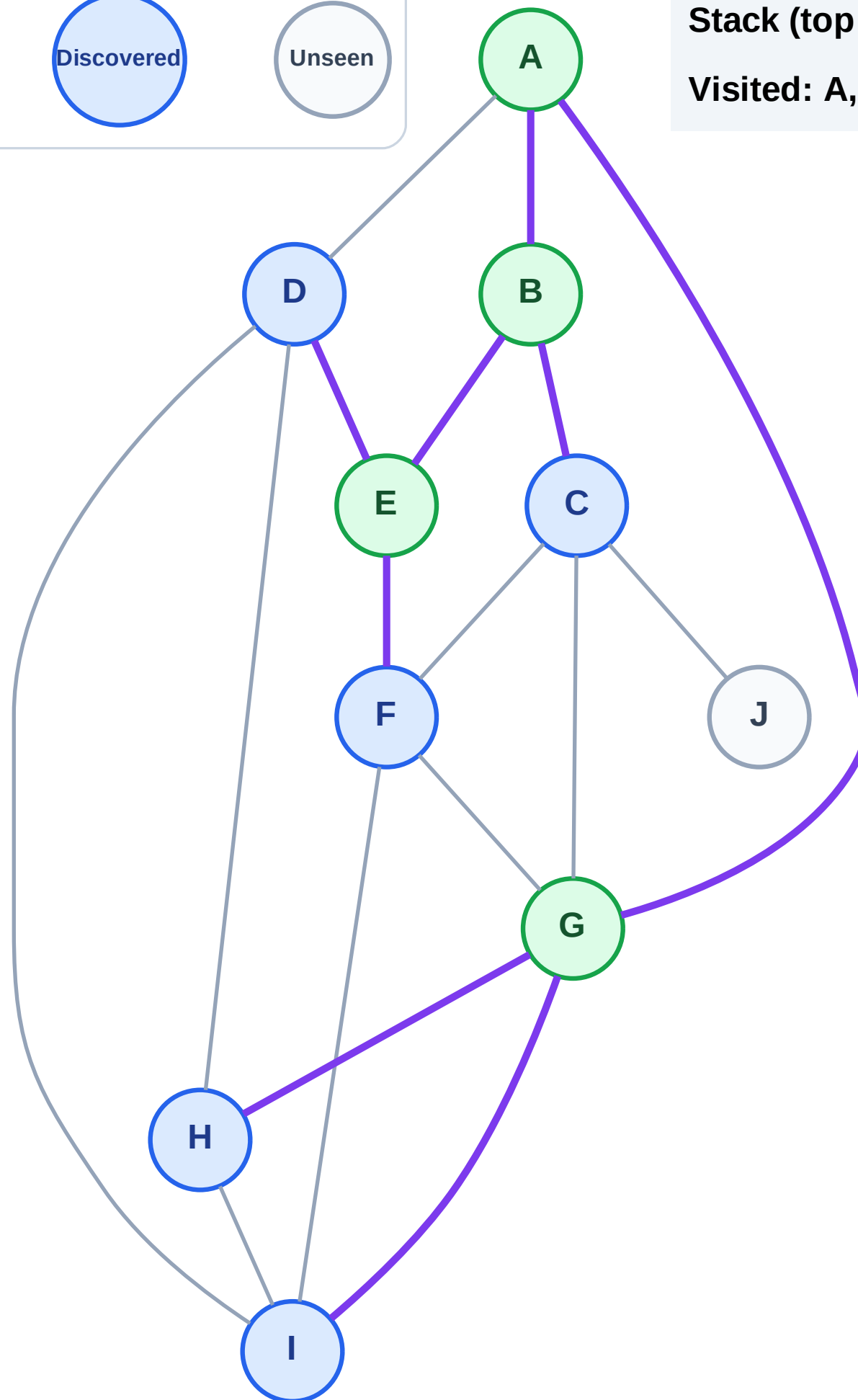
Discovered



Unseen

Stack (top at right): F | D | C | I | H

Visited: A, B, E, G



DFS Traversal

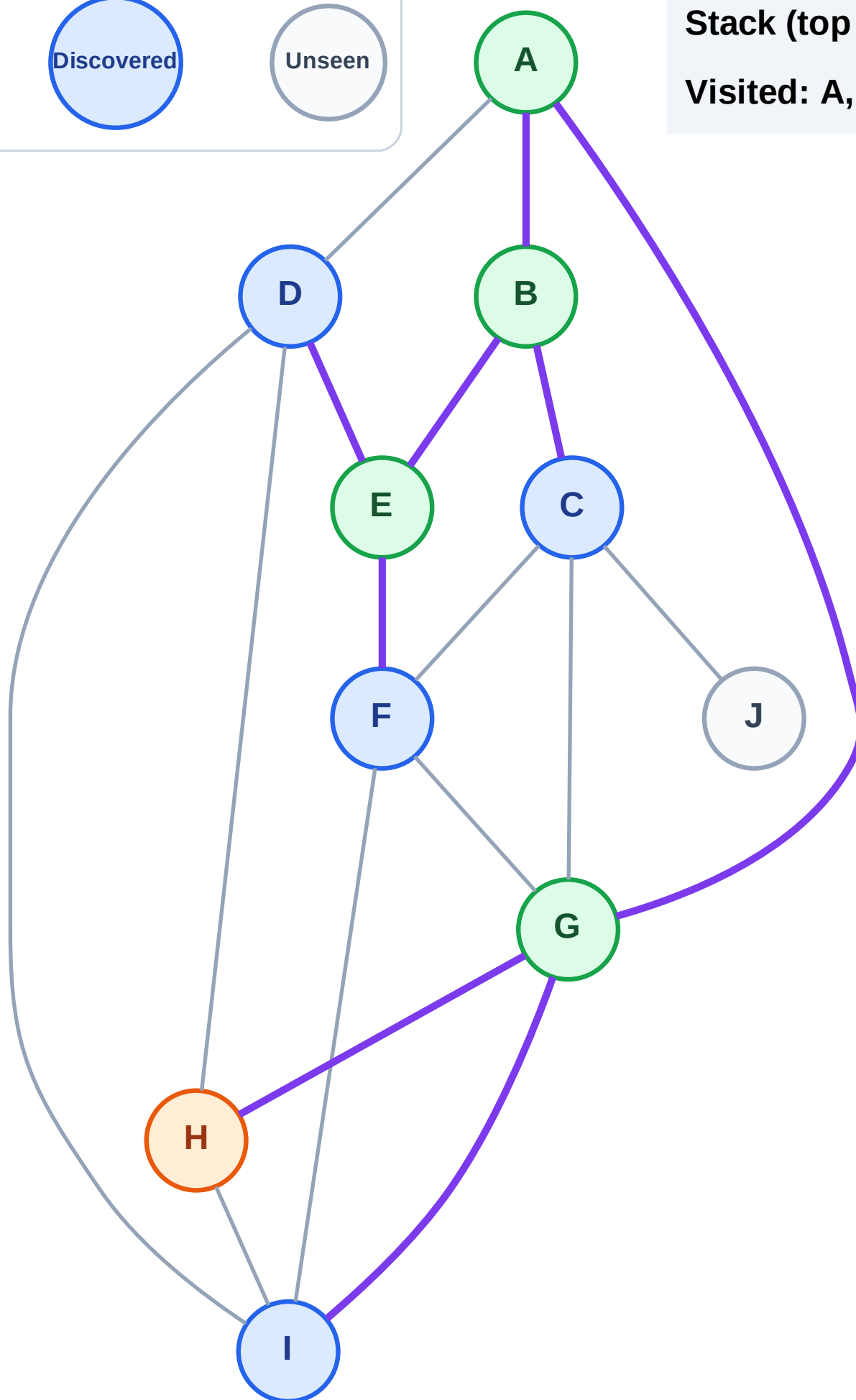
Step 10: Process node H

Legend



Stack (top at right): F | D | C | I

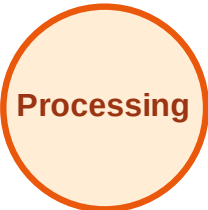
Visited: A, B, E, G



DFS Traversal

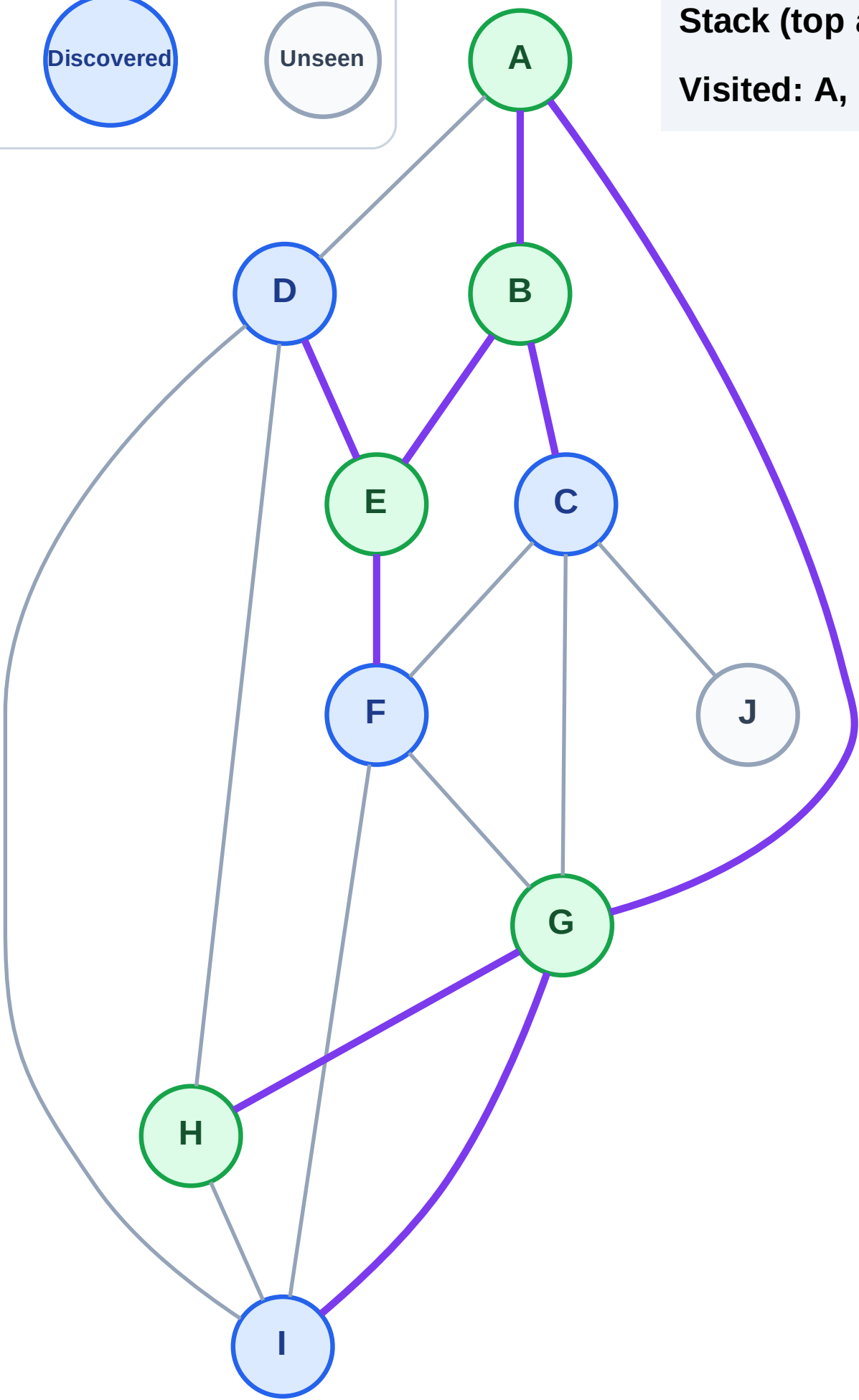
Step 11: No new neighbors from H

Legend



Stack (top at right): F | D | C | I

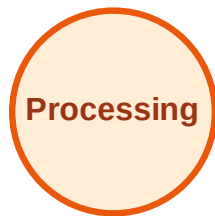
Visited: A, B, E, G, H



DFS Traversal

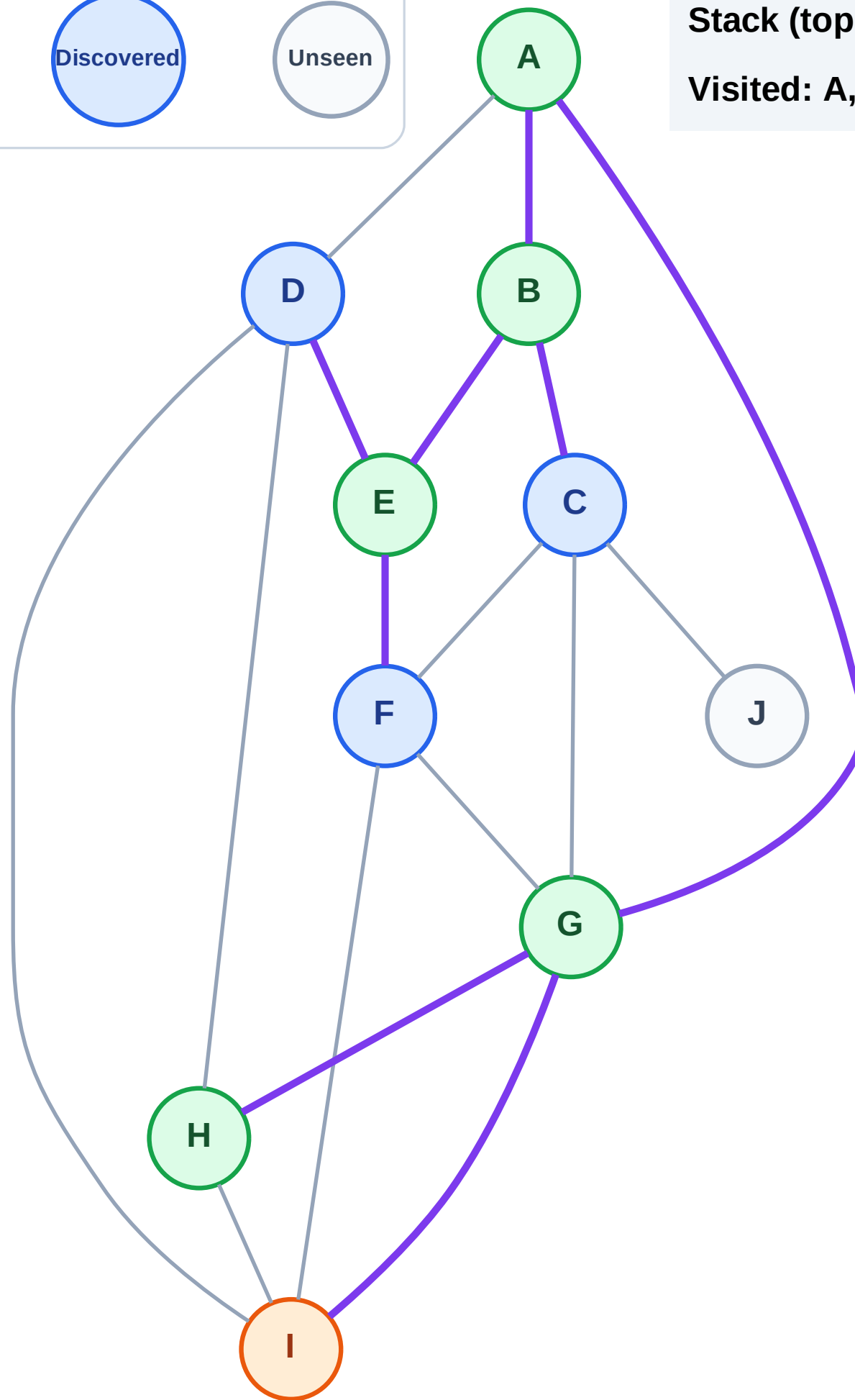
Step 12: Process node I

Legend



Stack (top at right): F | D | C

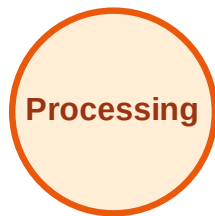
Visited: A, B, E, G, H



DFS Traversal

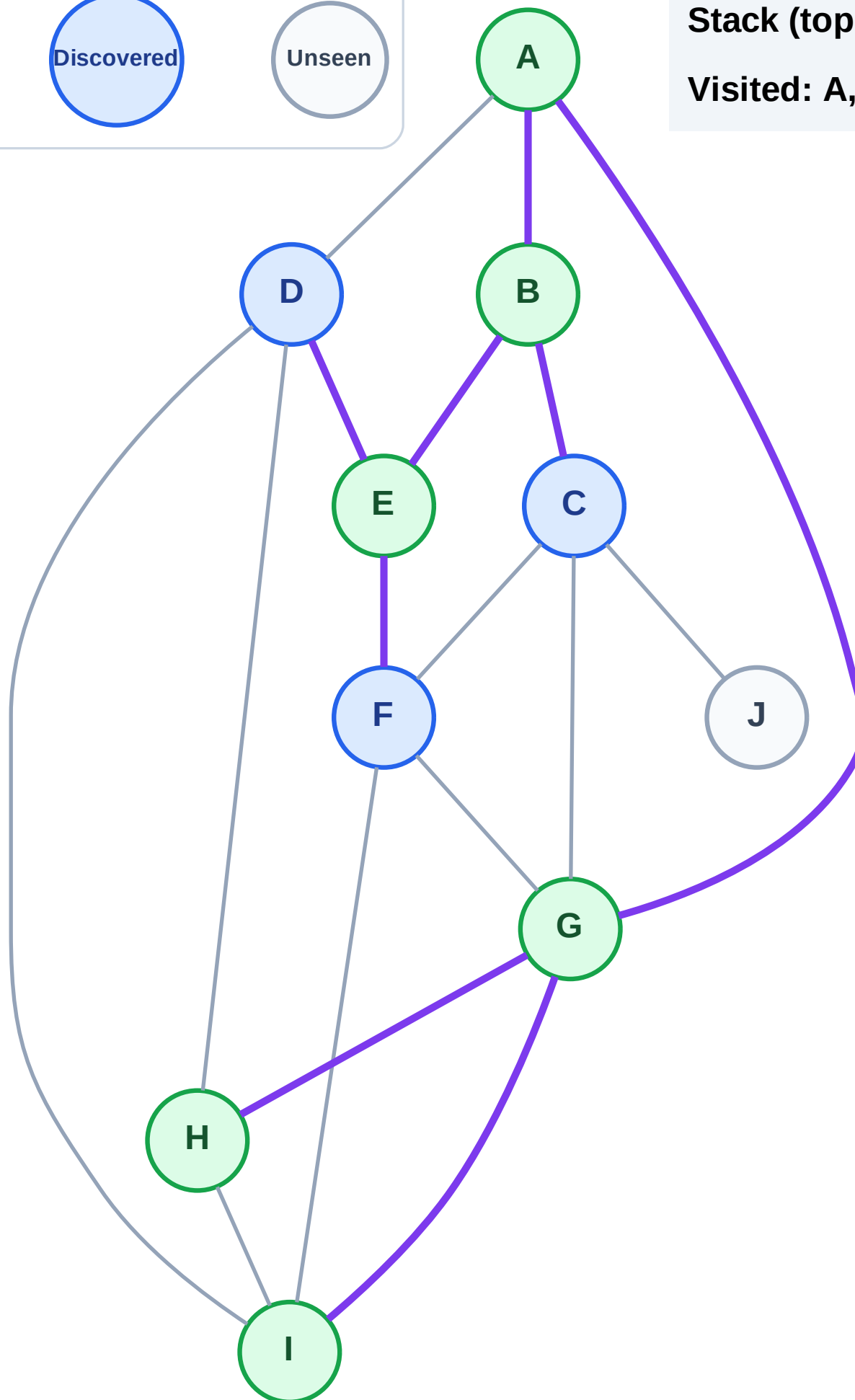
Step 13: No new neighbors from I

Legend



Stack (top at right): F | D | C

Visited: A, B, E, G, H, I



DFS Traversal

Step 14: Process node C

Legend

Complete

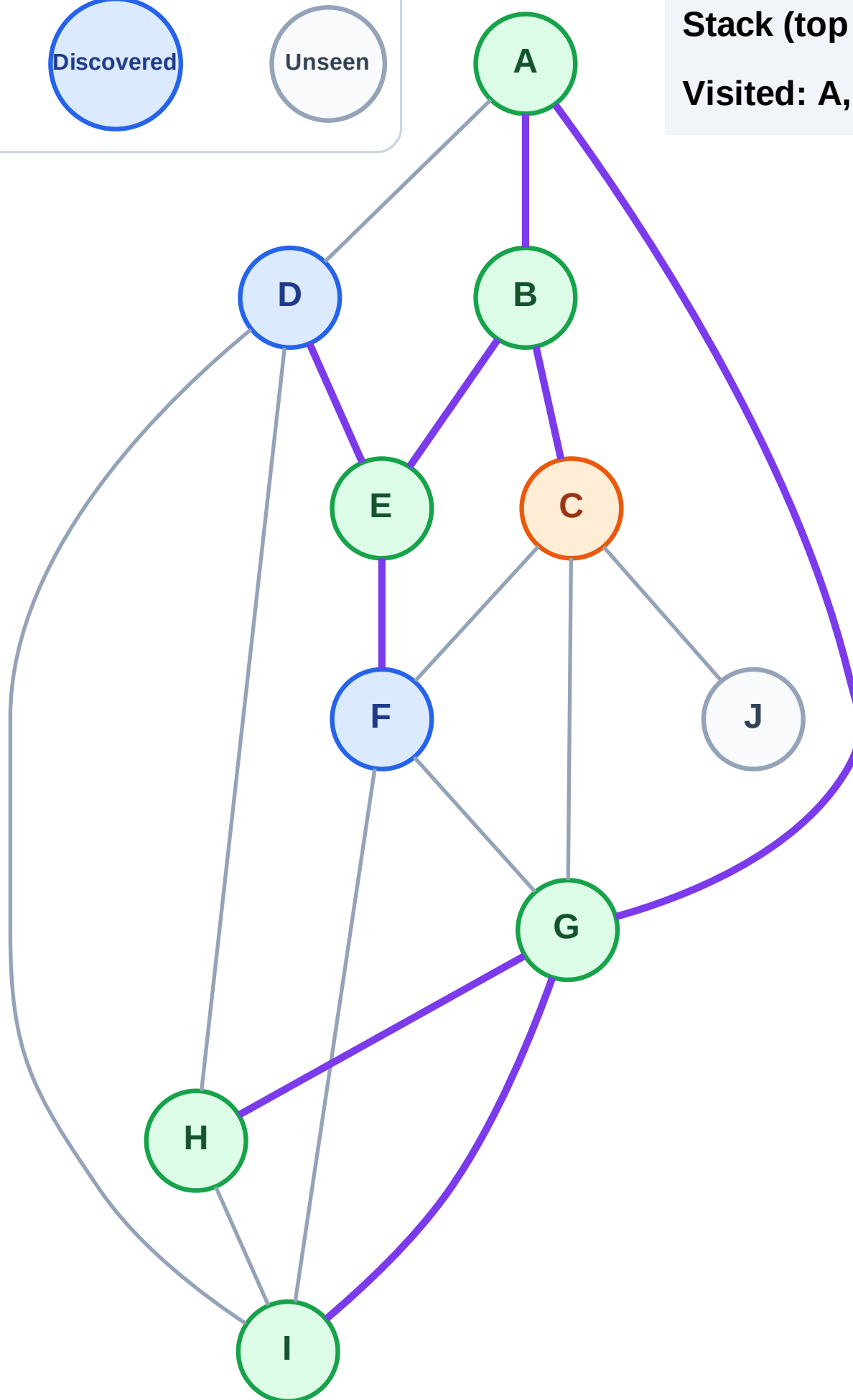
Processing

Discovered

Unseen

Stack (top at right): F | D

Visited: A, B, E, G, H, I



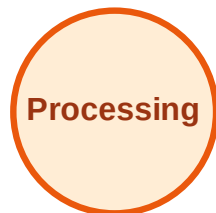
DFS Traversal

Step 15: Discover J from C

Legend



Complete



Processing



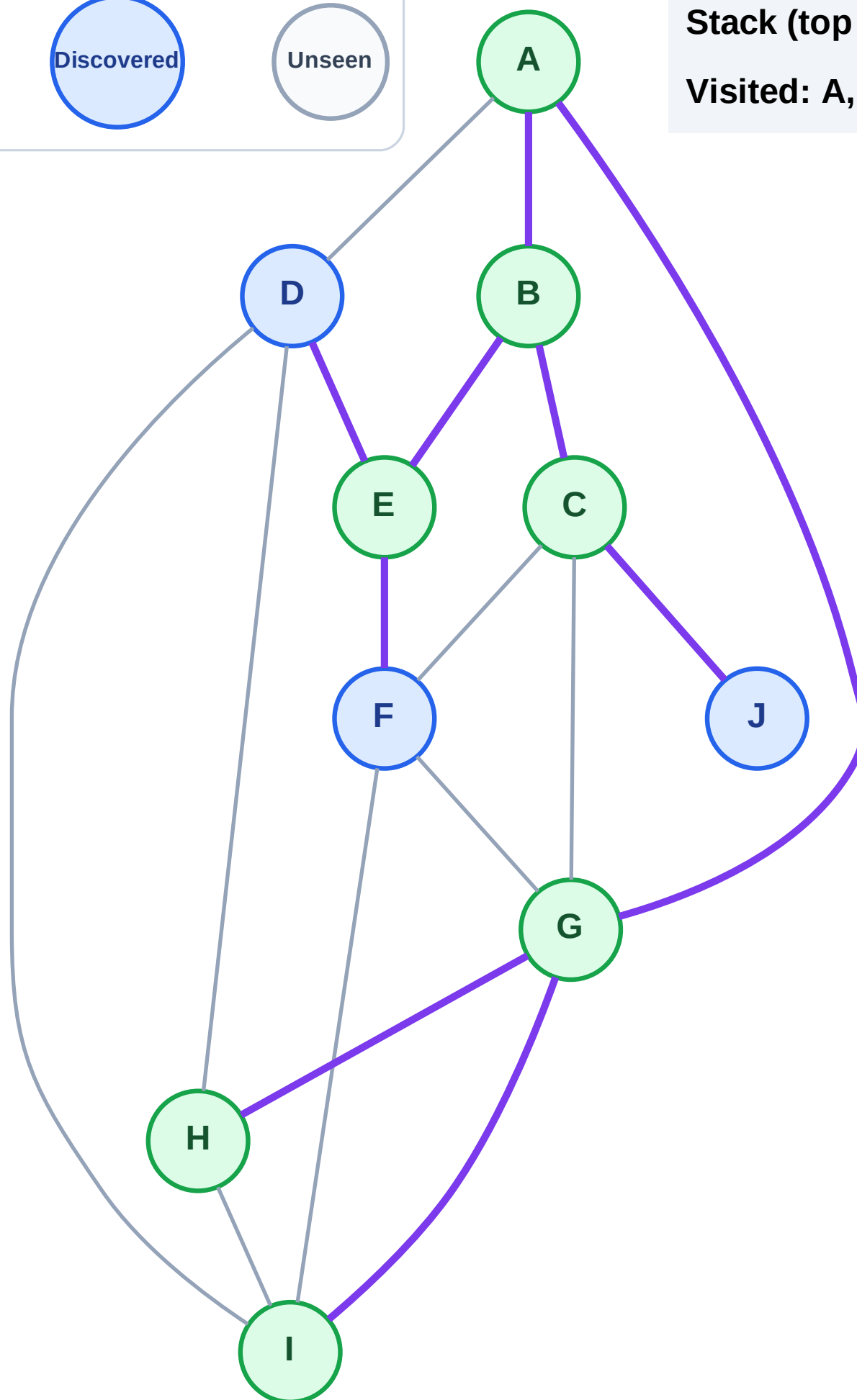
Discovered



Unseen

Stack (top at right): F | D | J

Visited: A, B, C, E, G, H, I



DFS Traversal

Step 16: Process node J

Legend

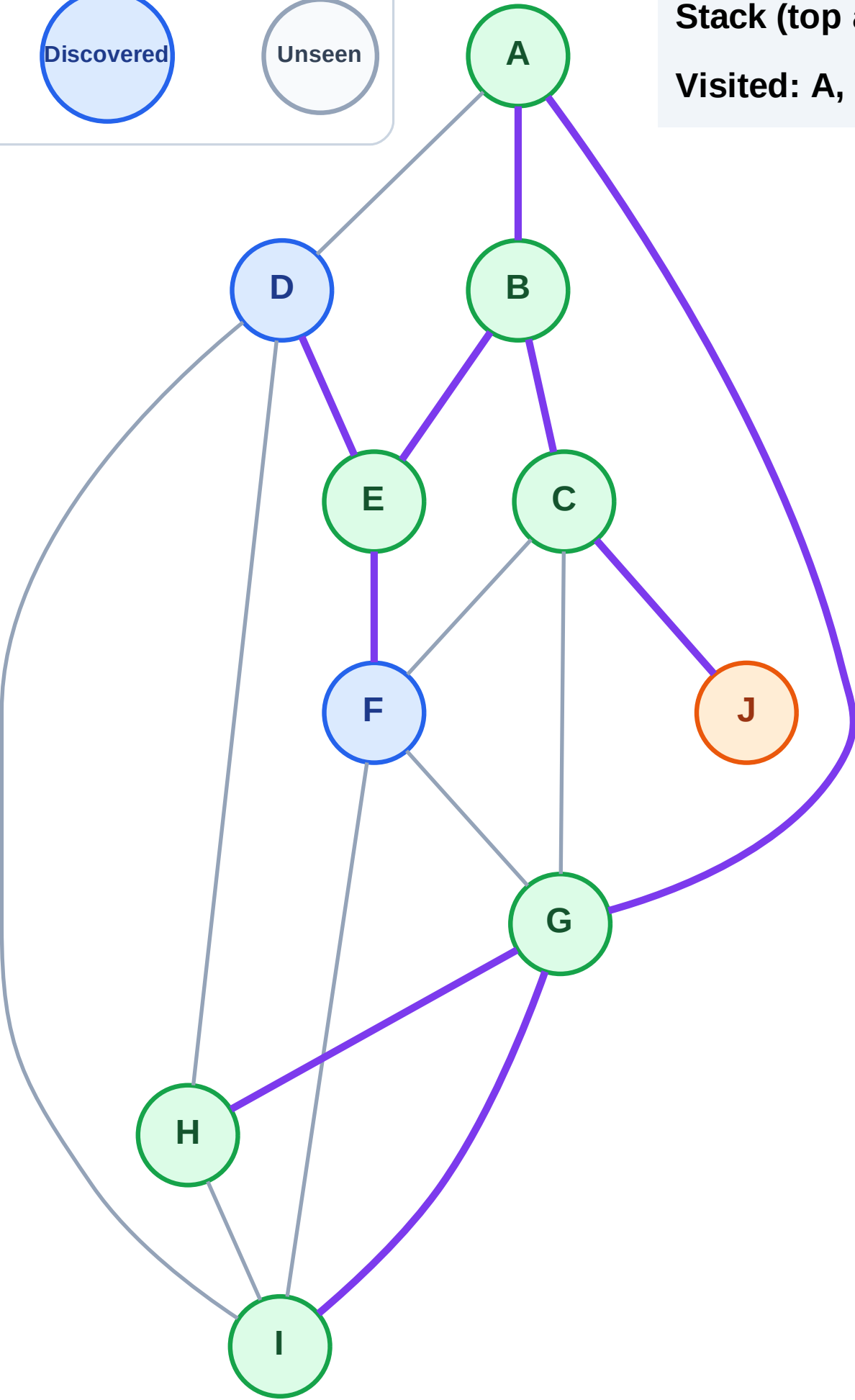
Complete

Processing

Discovered

Unseen

Stack (top at right): F | D
Visited: A, B, C, E, G, H, I



DFS Traversal

Step 17: No new neighbors from J

Legend

Complete

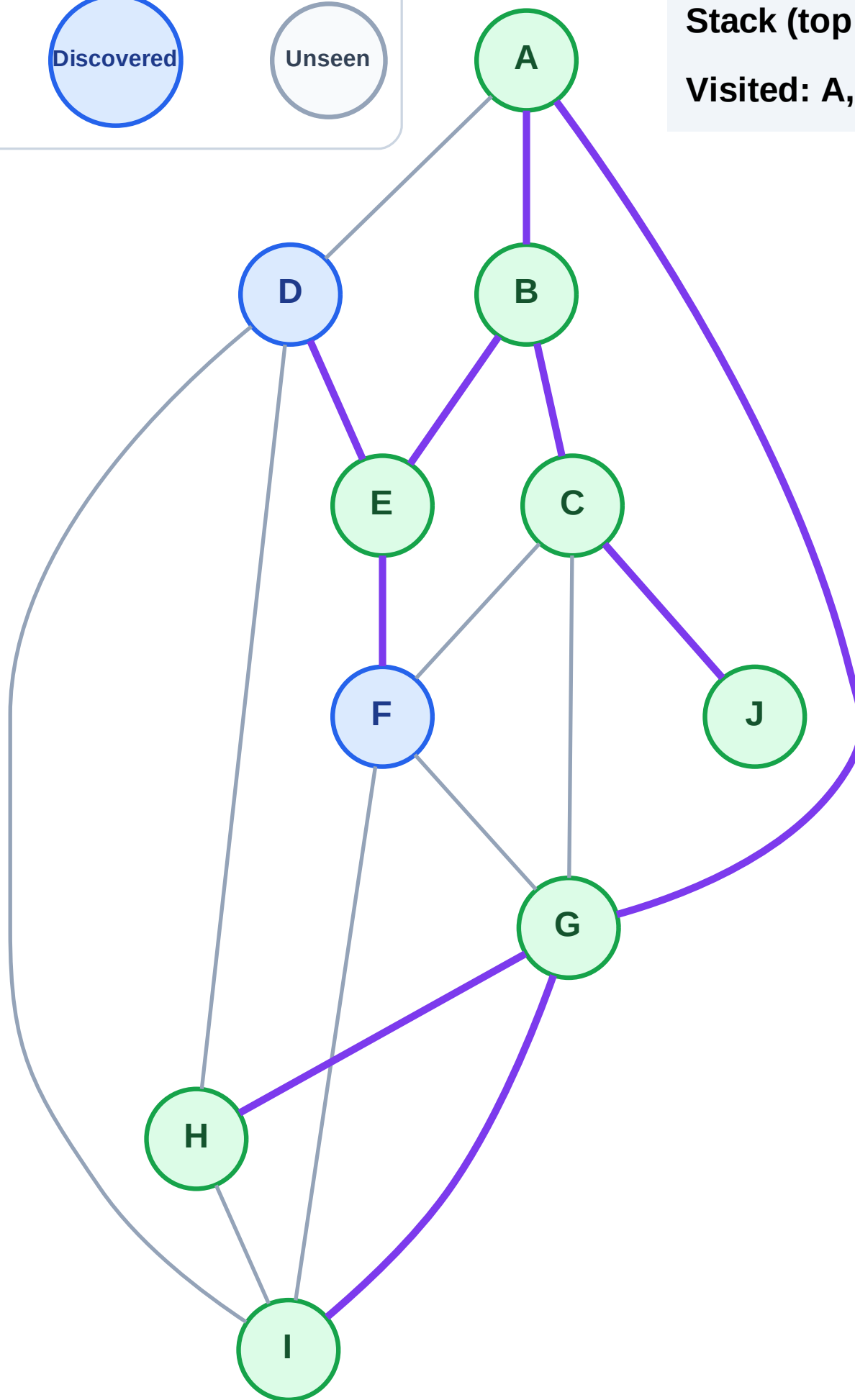
Processing

Discovered

Unseen

Stack (top at right): F | D

Visited: A, B, C, E, G, H, I, J



DFS Traversal

Step 18: Process node D

Legend

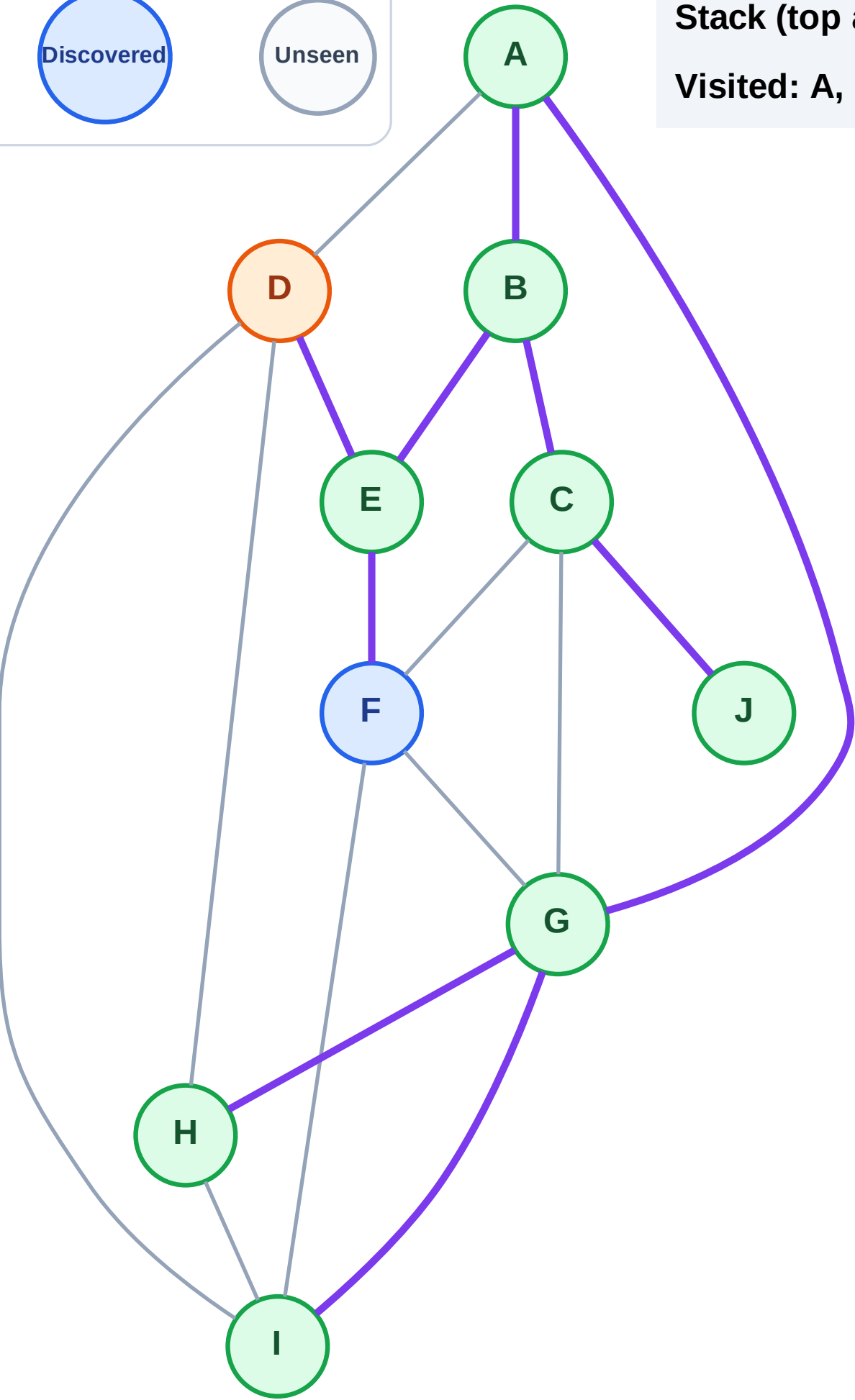
Complete

Processing

Discovered

Unseen

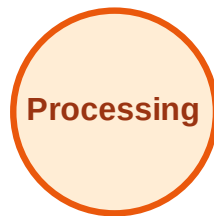
Stack (top at right): F
Visited: A, B, C, E, G, H, I, J



DFS Traversal

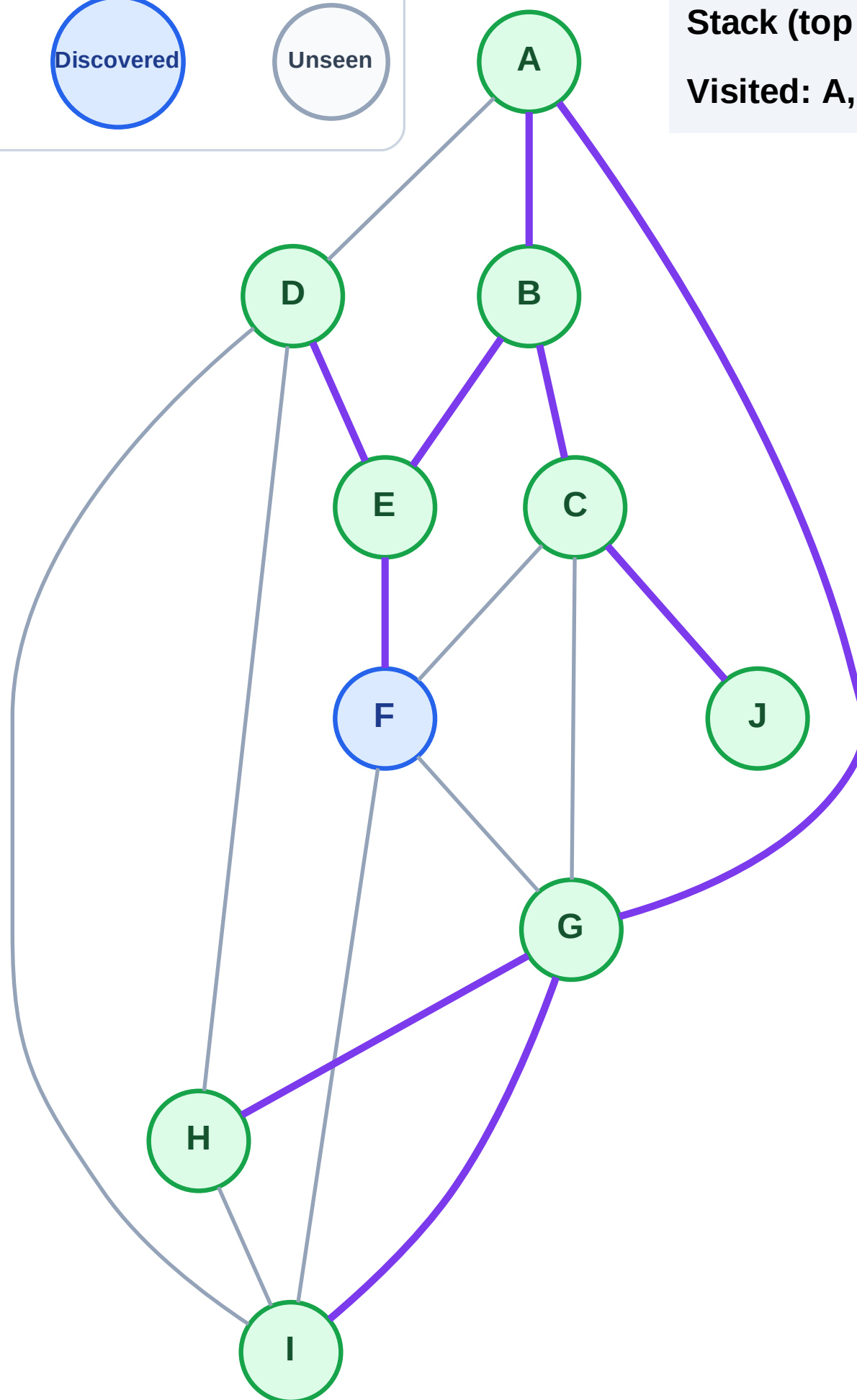
Step 19: No new neighbors from D

Legend



Stack (top at right): F

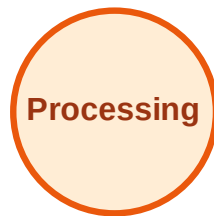
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

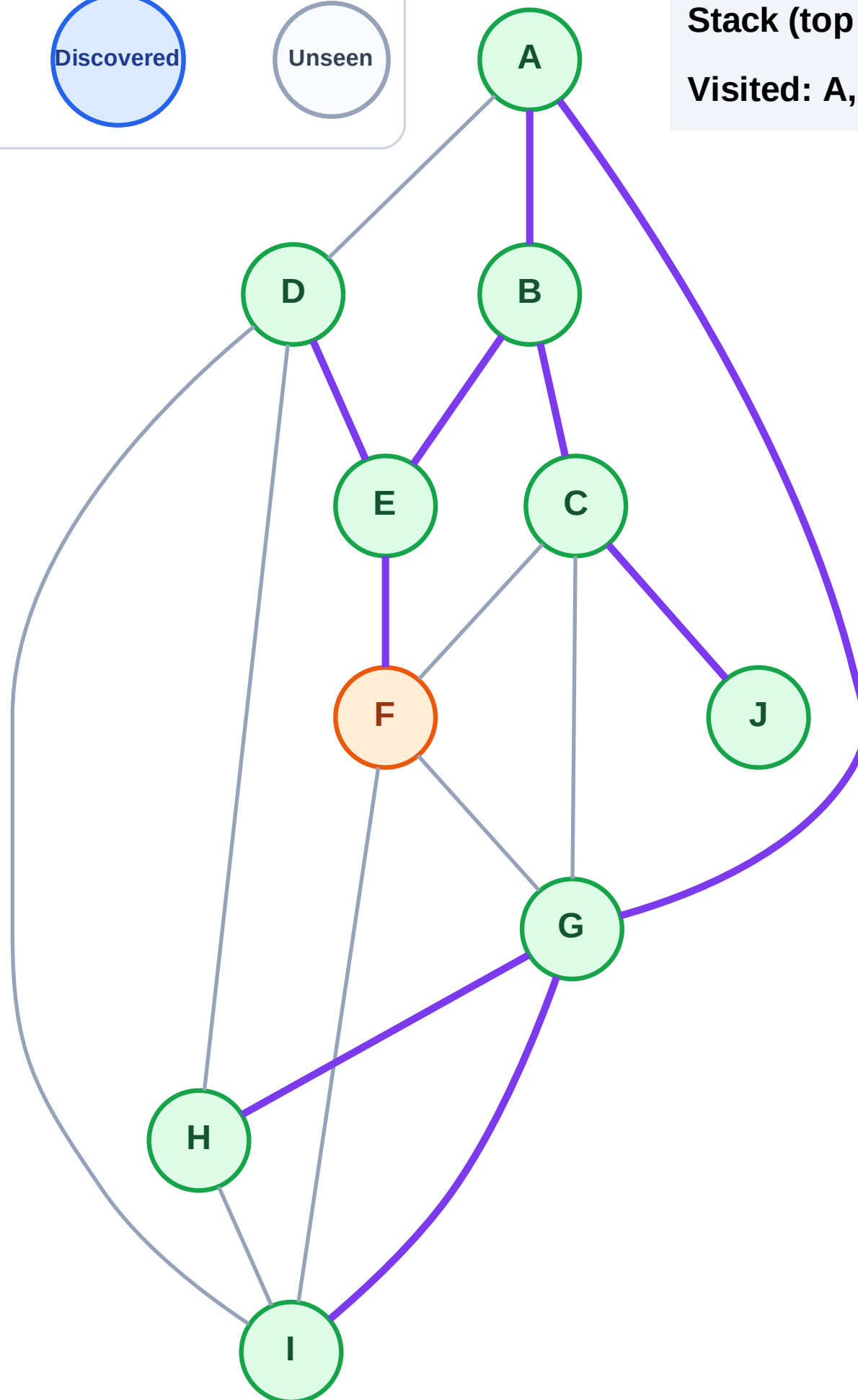
Step 20: Process node F

Legend



Stack (top at right): (empty)

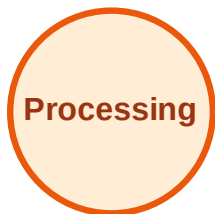
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

